Groups

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Contents

Lecture 1 – 10/10/25

§1 Examples and definitions

One can think about groups in two ways - *algebra* and *symmetry*.
We also include the *identity* symmetry of doing nothing.

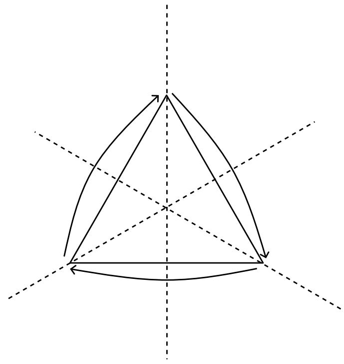

Figure 1: Symmetries of the equilateral triangle.

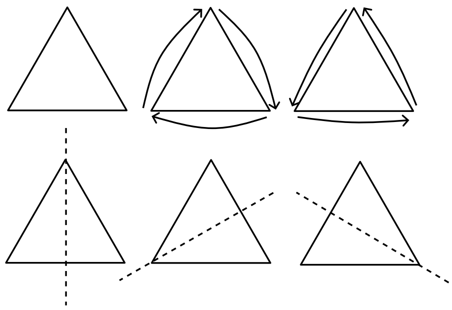

Figure 2: All 6 symmetries of an equilateral triangle.

> **Remark.** Can we generalise? How many symmetries does a regular n-gon have?

Things get more interesting if we *compose* – *multiply?* our symmetries.

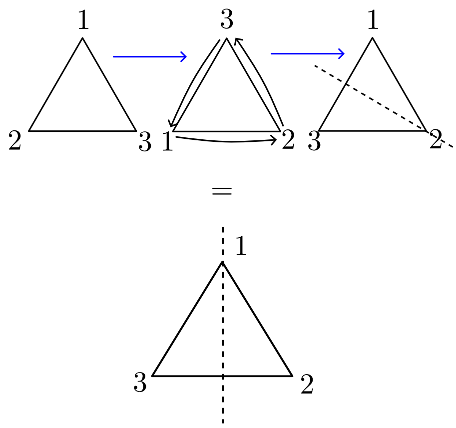

Figure 3: Composing our symmetries gives another symmetry!

Let's tabulate our properties.

- There is an *identity* symmetry.
- Every symmetry has an *inverse*.
  - The inverse of the identity is itself.
  - The inverse of a clockwise rotation is the corresponding counter-clockwise rotation.

- The inverse of a reflection is itself!

- It is possible to compose symmetries at all!
- *Associativity*: Composition of symmetries is associative – it doesn't matter how we bracket when we compose our symmetries.
- We *don't* necessarily have *commutativity*.

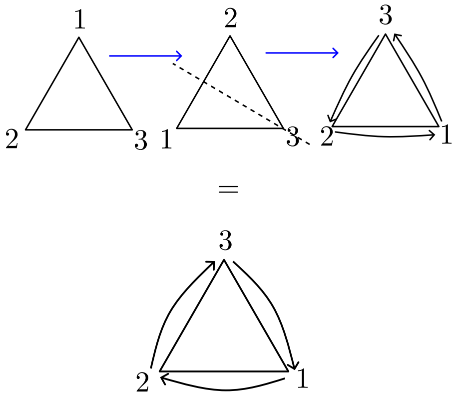

Figure 4: When we perform the operations from fig. 3 in the opposite order, the outcome is different.

Time to go back to algebra.

Definition 1.1. A *binary operation* on a set X is a function

$$\begin{aligned} \cdot : X \times X &\to X \\ (x, y) &\mapsto x \cdot y \end{aligned}$$

Definition 1.2. A *group* is a triple $(G, \cdot, e)$ where

- G is a set;
- $\cdot$ is a binary operation;
- and e is an element of G.

such that the following *group axioms* hold:

- *Closure*. For all a, $b \in G$, $a \cdot b \in G$ (we actually already know this because it's part of the definition of a binary operation).
- *Associativity*. For all a, b, $c \in G$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
- *Identity*. For all $a \in G$, $a \cdot e = a$.
- *Right inverses*. For all $a \in G$, there exists $b \in G$ such that $a \cdot b = e$.

Example

The symmetries of an equilateral triangle form a group.

Exercise 1.3. Show that $(\mathbb{Z}, +, 0)$ is a group.

Proposition 1.4

Let $(G, \cdot, e)$ be a group, and let a, b, b', e' be elements of G.

1. If $a \cdot b = e$, then $b \cdot a = e$ – right inverses are left inverses;
2. $e \cdot a = a$ – the identity is also a left identity;
3. If a, b and b' are such that $a \cdot b = a \cdot b' = e$, then $b = b'$ – inverses are unique;
4. If $a \cdot e' = a$ then $e' = e$ – the identity is unique.

Lecture 2 – 13/10/25

Proposition 1.5

Let $(G, \cdot, e)$ be a group, and let a, b, b', e' be elements of G.

1. If $a \cdot b = e$, then $b \cdot a = e$ – right inverses are left inverses;
2. $e \cdot a = a$ – the identity is also a left identity;
3. If a, b and b' are such that $a \cdot b = a \cdot b' = e$, then $b = b'$ – inverses are unique;
4. If $a \cdot e' = a$ then $e' = e$ – the identity is unique.

Proof. We simply prove each part sequentially.

1. Suppose $a \cdot b = e$. Then

$$\begin{aligned}
b &= b \cdot e && \text{by identity} \\
&= b \cdot (a \cdot b) && \text{by assumption} \\
&= (b \cdot a) \cdot b && \text{by associativity} \\
\implies b \cdot b^{-1} &= ((b \cdot a) \cdot b) \cdot b^{-1} && \text{where } b^{-1} \text{ is an inverse of } b \\
&= (b \cdot a) \cdot (b \cdot b^{-1}) && \text{by associativity} \\
\implies e &= (b \cdot a) \cdot e && \text{by definition of the inverse} \\
&= b \cdot a && \text{by definition of the identity}
\end{aligned}$$

2. Let a^{-1} be an inverse of a, then by 1, $a \cdot a^{-1} = e = a^{-1} \cdot a$. Now

$$\begin{aligned}
e \cdot a &= (a \cdot a^{-1}) \cdot a \\
&= a \cdot (a^{-1} \cdot a) \\
&= a \cdot e \\
&= a
\end{aligned}$$

3. Suppose $a \cdot b = a \cdot b' = e$. Then

$$b' = e \cdot b' \qquad \text{by 2}$$

$$
\begin{aligned}
&= (b \cdot a) \cdot b' && \text{by } 1\\
&= b \cdot (a \cdot b')\\
&= b \cdot e\\
&= b
\end{aligned}
$$

4. Let a^{-1} be the inverse of a, so $a^{-1} \cdot a = e$.

$$
\begin{aligned}
a &= a \cdot e'\\
\implies a^{-1} \cdot a &= a^{-1} \cdot (a \cdot e')\\
&= (a^{-1} \cdot a) \cdot e'\\
&= e \cdot e'\\
&= e'
\end{aligned}
$$

Hence $e = e'$. Note we only needed e' to be a 'second inverse' for one element!

■

A corollary of the uniqueness of inverses along with right inverses also being left inverses is that a is the inverse of a^{-1}.

Definition 1.6 (Powers of elements)**.** Fix $a \in G$, then

- $a^0 = e$;
- For $n \in \mathbb{N}$, $a^n = a^{n-1} \cdot a$.
- For $n < 0$, $a^n = (a^{-1})^{-n}$.

Proposition 1.7

For $a \in G$, $n, m \in \mathbb{Z}$,

- $a^m \cdot a^n = a^{m+n}$;
- $(a^m)^n = a^{mn}$.

Proposition 1.8

For $a, b \in G$, $(a \cdot b)^{-1} = b^{-1} \cdot a^{-1}$.

Definition 1.9 (Abelian groups)**.** For a group $(G, \cdot, e)$, if $a \cdot b = b \cdot a$ for all $a, b \in G$, then G is *abelian*.

Example (The trivial group)

The *trivial group* has $G = \{e\}$ and $e \cdot e = e$

Example (Additive groups)
$(\mathbb{Z}, +, 0)$, $(\mathbb{Q}, +, 0)$, $(\mathbb{R}, +, 0)$, $(\mathbb{C}, +, 0)$ – all abelian. $(\mathbb{N}, +, 0)$ is *not* a group.

Example (Multiplicative groups)
$(\mathbb{Q}, \times, 1)$ is not a group – but $(\mathbb{Q} \setminus \{0\}, \times, 1)$ is (abelian). As are $(\mathbb{R} \setminus \{0\}, \times, 1)$ and $(\mathbb{C} \setminus \{0\}, \times, 1)$.

Definition 1.10. The *order* of a group $(G, \cdot, e)$ is the number of elements of G, $|G|$. If $|G| < \infty$, then $(G, \cdot, e)$ is *finite*.

Example (Finite groups)
- For $n \in \mathbb{N}$, let $C_n = \{z : z \in \mathbb{C}, z^n = 1\}$. Then $(C_n, \times, 1)$ is an abelian group.
- For $n \in \mathbb{N}$, let $Z_n = \{0, \ldots, n-1\}$. Let $a +_n b$ be the remainder when $a + b$ is divided by n. Then $(Z_n, +_n, 0)$ is an abelian group.

Lecture 3 – 15/10/25

§1.1 Symmetric groups

Definition 1.11. Let X, Y be sets. A *bijection* is a map $f : X \to Y$ that has an inverse $g : Y \to X$ such that $f \circ g = \mathrm{id}_Y$ and $g \circ f = \mathrm{id}_X$.

Definition 1.12. A bijection from a set X to itself is called a *permutation*.

Definition 1.13. Given a set X, the set $\mathrm{Sym}(X)$ is the set of permutations of the set.

Lemma 1.14 (Associativity of function composition)
Consider maps of sets:

$$W \xrightarrow{f} X \xrightarrow{g} Y \xrightarrow{h} Z$$

Then $(h \circ g) \circ f = h \circ (g \circ f)$.

Proof. For any $w \in W$, $((h \circ g) \circ f)(w) = h(g(f(w))) = (h \circ (g \circ f))(w)$. ∎

And that's all we need to show to make our next group!

Proposition 1.15
For any set X, $(\mathrm{Sym}(X), \circ, \mathrm{id}_X)$ is a group.

Proof. We must verify the four group axioms.

- *Closure*. $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$, so the composition of two permutations is a permutation.
- *Associativity*. By lemma 1.14.
- *Identity*. id_X.
- *Inverse*. By definition: we are working with bijections.

■

Definition 1.16. If $X = \{1, \ldots, n\}$, we write S_n for $(\mathrm{Sym}(X), \circ, \mathrm{id}_X)$.

Example

- S_3, for example, is the group of ways to arrange 3 flowerpots on the windowsill.
- S_{52} is the group of ways to shuffle a deck of cards.

Fairly clear that $|S_n| = n!$.

> **Remark** (Notation)**.** Writing $(G, \cdot, e)$ is cumbersome. Let's just write G and leave the rest as implied. If we really want to emphasise that e is the identity, we can always write e_G, and $\cdot_G$ for $\cdot$. Finally we might be so apathetic that we even write ab instead of $a \cdot b$.

§1.2 Subgroups

We might want to focus on smaller groups within a group – for example, $\mathbb{Z}$ inside $\mathbb{R}$, or the rotations of a triangle instead of all its symmetries.

Definition 1.17. Let G be a group, and $H \subseteq G$. If H satisfies

- $e \in H$;
- for any a, $b \in H$, $ab \in H$;
- for any $a \in H$, $a^{-1} \in H$

then we say H is a *subgroup* of G. We write $H \leq G$.

> **Remark.** If G is a group and H is a subgroup, then H is itself a group.

Example

- Every group G has itself and $\{e\}$ as subgroups. Any other subgroup is called *proper*.
- $\mathbb{Z} \leq \mathbb{Q} \leq \mathbb{R} \leq \mathbb{C}$.
- For $n \in \mathbb{N}_0$, let $n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$ – the set of multiples of n. Then $n\mathbb{Z} \leq \mathbb{Z}$. In fact these are the only subgroups of $\mathbb{Z}$

Proposition 1.18

If $H \leq \mathbb{Z}$, $H = n\mathbb{Z}$ for some $n \in \mathbb{N}_0$.

Proof. If $H = \{0\}$, then $H = 0\mathbb{Z}$. If $H \neq \{0\}$, then let n be the smallest positive integer in H. By induction, $nk \in H$ for all $k \in \mathbb{Z}$, so $n\mathbb{Z} \leq H$. Now suppose for contradiction there exists $h \in H$ such that $h \notin n\mathbb{Z}$. Dividing h by n gives $h = nq + r$ with $0 \leq r < n$, and in particular by closure $r \in H$. Furthermore, $r \neq 0$ since $h \notin n\mathbb{Z}$. But this contradicts the minimality of n. Hence $H = n\mathbb{Z}$ as required. ∎

Lecture 4 – 17/10/25

Proposition 1.19

If $H, K \leq G$, then $H \cap K \leq G$

Proof. (Example sheet 1) ∎

> **Remark.** We can generalise to any collection of subgroups.

Definition 1.20. Let X be a subset of a group G. Then

$$\langle X \rangle := \bigcap_{X \leq H \leq G} H$$

is the *subgroup generated by* X – the intersection of all subgroups of G that contain X. Intuition: the smallest subgroup of G containing X.

Definition 1.21. If $\langle X \rangle = G$, we say that X *generates* G or is a *generating set* for G. If X generates G, it means that every $g \in G$ can be written as

$$x_1^{\pm 1} \cdot x_2^{\pm 1} \cdots x_n^{\pm 1}$$

for some $n \geq 0$, where every $x_i \in X$.

§1.3 Geometric examples

Let $\mathbb{C}$ be the plane, equipped with the usual notion of distance.

DIAGRAM: complex plane, w = w1 + iw2, z = z1 + iz2, line between them, labelled —z-w— = an expression due to Pythagoras

Definition 1.22. For any subset $X \subseteq \mathbb{C}$, an *isometry* of X is a bijection f from X to itself that preserves distance – for any $x, y \in X$, $|f(x) - f(y)| = |x - y|$.

Proposition 1.23 (Isometry groups in $\mathbb{C}$)

Let $X \subseteq \mathbb{C}$. The set of isometries of X, $\mathrm{Isom}(X)$, is a subgroup of $\mathrm{Sym}(X)$. In particular, $\mathrm{Isom}(X)$ is a group.

Proof. Let f, $g \in \mathrm{Isom}(X)$, and x, $y \in X$.

- *Identity*. Clearly id_X is an isometry, so it is in $\mathrm{Isom}(X)$ as required.
- *Closure*. Clear that the composition of isometries is an isometry by the transitivity of equality.
- *Inverse*. If f is an isometry,

$$\begin{aligned} |f^{-1}(x) - f^{-1}(y)| &= |f \circ f^{-1}(x) - f \circ f^{-1}(y)| \\ &= |x - y| \end{aligned}$$

∎

Definition 1.24. Let $n \geq 3$ and let $X_n \subseteq \mathbb{C}$ be the n-gon with vertices $\{\exp(\frac{2\pi ik}{n}) : k = 0, \dots, n-1\}$.

DIAGRAM: 6 points equally spaced around the unit circle in the complex plane, one point at $n = 1$, label 'e.g. $n = 6$', hexagon drawn.

Then the n^{th} *dihedral group*, $D_{2n} := \mathrm{Isom}(X_n)$. So for example, D_6 is the symmetry group of an equilateral triangle.

Theorem 1.25

The size of the n^{th} dihedral group is $2n$.

First, we need some geometric lemmas.

Lemma 1.26 (Kite lemma)

Let x_1, x_2, y_1, $y_2 \in \mathbb{C}$. If $|y_1 - x_1| = |y_2 - x_1$ and $|y_1 - x_2| = |y_2 - x_2|$, then $x_2 - x_1$ is perpendicular to $y_2 - y_1$.

Proof. DIAGRAM: Kite: diagonals y1y2 and x1x2. Whimsical kite tail attached to x2. Equal lengths of x1y1 and x1y2 and x2y1 and x2y2 indicated. Diagonals drawn. Point z labelled at intersection.

By symmetry, $\angle X_1ZY_1 = \angle X_1ZY_2$. But Y_1Y_2 is a straight line, so $\angle X_1ZY_1 + \angle X_1ZY_2 = \pi$. Hence $\angle X_1ZY_1 = \angle X_1ZY_2 = \frac{\pi}{2}$ as required. ∎

Lemma 1.27 (Three point lemma)

Let $X \subseteq \mathbb{C}$ and $f \in \mathrm{Isom}(X)$. Suppose x_1, x_2, $x_3 \in X$ distinct and not collinear, such that $f(x_i) = x_i$ for $i = 1, 2, 3$. Then $f = \mathrm{id}_X$.

Proof. Suppose $f(y) \neq y$ for some $y \in X$. Then

$$\begin{aligned} |f(y) - x_i| &= |f(y) - f(x_i)| \\ &= |y - x_i| && \text{since } f \in \mathrm{Isom}(X) \end{aligned}$$

Now apply the kite lemma with $y_1 = y$, $y_2 = f(y)$ and letting x_1, x_2 range over the x_i to get $f(y) - y$ perpendicular to $x_i - x_j$ for any $i \neq j$. Then, since $f(y) - y \neq 0$, $x_i - x_j$ are all parallel, so x_1, x_2, x_3 are collinear.

DIAGRAM: ????

■

> **Remark.** There is equally an $(n+1)$-point lemma valid in $\mathbb{R}^n$.

Lecture 5 – 20/10/25

Proof of theorem. We first define two particular elements of D_{2n}:

$$r : z \mapsto e^{\frac{2\pi i}{n}} z \qquad \text{rotation}$$
$$s : z \mapsto \overline{z} \qquad \text{reflection}$$

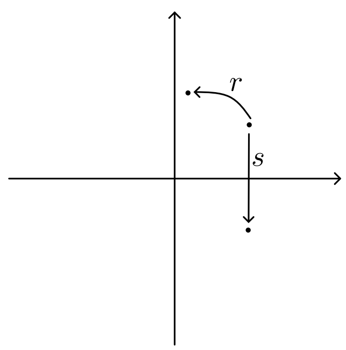

Figure 5: Graphical depictions of r and s

We will now further claim that $D_{2n} = \{e, r, \ldots, r^{n-1}, s, rs, \ldots, r^{n-1}s\}$. In particular, $\{r, s\}$ generates D_{2n}.

First, it is necessary to show that r, $s \in D_{2n}$. For some x, $y \in \mathbb{C}$,

- $|r(x) - r(y)| = |e^{\frac{2\pi i}{n}} x - e^{\frac{2\pi i}{n}} y| = |e^{\frac{2\pi i}{n}}||x - y| = |x - y|$, so r is an isometry. Furthermore, $r(e^{\frac{2\pi i k}{n}}) = e^{\frac{2\pi i(k+1)}{n}}$, so r sends vertices to other vertices, so r really is a permutation of X_n.
- $|s(x) - s(y)| = |\overline{x} - \overline{y}| = |\overline{x - y}| = |x - y|$, so s is an isometry. To see that s preserves the vertices of X_n, we can either proceed algebraically or appeal to the symmetry of X_n in the real axis.

From this it follows that $\{e, r, \ldots, r^{n-1}, s, rs, \ldots, r^{n-1}s\} \subseteq D_{2n}$ by closure. It remains to show that these are all the elements. Let f arbitrary in D_{2n}. Let $x = 1$, $y = e^{\frac{2\pi i}{n}}$, $z = e^{-\frac{2\pi i}{n}}$ (DIAGRAM?). Since $f \in D_{2n}$, $f(x)$ is some vertex of X_n – that is, $f(x) = e^{\frac{2\pi i k}{n}}$ for some $k \in \{0, \ldots, n-1\}$. Then $r^{-k} \circ f(x) = 1$. Since $r^{-k} \circ f$ is an isometry, we can also see $r^{-k} \circ f(y)$ and $r^{-k} \circ f(z)$ are themselves y and z in some order.

1. $r^{-k} \circ f(y) = y$ and $r^{-k} \circ f(z) = z$. Then, by the 3 point lemma, $r^{-k} \circ f = e$. So $f = r^k \in \{e, r, \ldots, r^{n-1}, s, rs, \ldots, r^{n-1}s\}$.

2. $r^{-k} \circ f(y) = z$ and $r^{-k} \circ f(z) = y$. Then $s \circ r^{-k} \circ f(y) = y$ and $s \circ r^{-k} \circ f(z) = z$, so $s \circ r^{-k} \circ f = e$. Hence $f = r^k s$, since s is self-inverse, so $f \in \{e, r, \ldots, r^{n-1}, s, rs, \ldots, r^{n-1}s\}$, as required.

Thus $D_{2n} \subseteq \{e, r, \ldots, r^{n-1}, s, rs, \ldots, r^{n-1}s\}$. Finally, it is necessary to show that these elements are all distinct. One can simply verify that x and y are taken to different points by each of these. This concludes the proof.

■

To understand the group operation on D_{2n}, it is desirable to understand all the different ways to multiply r and s.

Lemma 1.28 (Dihedral relation)
For r, $s \in D_{2n}$ as above,

$$sr = r^{-1}s$$

Proof. One can appeal to intuition or just check for a small number of cases by the 3 point lemma. ■

Lecture 6 – 22/10/25

§1.4 Homomorphisms

Some groups are not set-theoretically equal, but nevertheless have the same structure – for example, Sym(deck of cards) and Sym(52 undergraduate mathematicians).

Definition 1.29 (Homomorphism)**.** A map between groups $\phi : G \to H$ is called a *homomorphism* if, for any g, $g' \in G$,

$$\phi(g \cdot g') = \phi(g) \cdot \phi(g')$$

Example
1. For any groups G, H, the map $\phi : G \to H$, $g \mapsto e_H$ is the *trivial homomorphism.*
2. If $H \leq G$, the map $i : H \to G$, $h \mapsto h$ is the *inclusion homomorphism*
3. If $nk = m$, the map $\phi : C_m \to C_n$, $g \mapsto g^k$ is a homomorphism.
4. Since $\det(AB) = \det(A)\det(B)$, $\det : \mathrm{GL}_2\,\mathbb{R} \to (\mathbb{R} \setminus \{0\}, \times)$, $A \mapsto \det A$ is a homomorphism.

Lemma 1.30
If $\phi : G \to H$ is a homomorphism, then

1. $\phi(e_G) = \phi(e_H)$, and
2. $\phi(g^{-1}) = \phi(g)^{-1}$ for all $g \in G$.

Proof.

1. $\phi(e_G) \cdot \phi(e_G) = \phi(e_G \cdot e_G) = \phi(e_G)$, so $\phi(e_G) = e_H$, by uniqueness of identities.
2. $\phi(g) \cdot \phi(g^{-1}) = \phi(g \cdot g^{-1}) = \phi(e_G) = e_H$, so $\phi(g)^{-1} = \phi(g^{-1})$, by uniqueness of inverses.

■

Definition 1.31 (Isomorphisms)**.** If a homomorphism $\phi : G \to H$ is also a bijection, then ϕ is called an *isomorphism*, and we write $G \cong H$.

Example

1. $\mathbb{Z}_n \cong C_n$ – we have the isomorphism $\phi : \mathbb{Z}_n \to C_n$, $k \mapsto e^{2\pi ik/n}$.
2. $(\mathbb{R}, +, 0) \cong (\mathbb{R}_{>0}, \times, 1)$ – we have the isomorphism $\phi : (\mathbb{R}, +, 0) \to (\mathbb{R}_{>0}, \times, 1)$, $x \mapsto \exp(x)$.

Lemma 1.32

1. If $\phi : G \to H$ is an isomorphism, so is ϕ^{-1}.
2. If $\phi : G \to H$ and $\psi : H \to K$ are homomorphisms, so is $\psi \circ \phi$.
3. $\cong$ is an equivalence relation.

Proof. Exercise. ■

We saw earlier that every subgroup induces an inclusion homomorphism. Furthermore, every homomorphism induces a subgroup.

Definition 1.33 (Image and kernel)**.** Let $\phi : G \to H$ be a homomorphism.

- The *image* of ϕ, $\operatorname{im}\phi$, is $\{h \in H : h = \phi(g) \text{ for some } g \in G\}$. (DIAGRAM)
- The *kernel* of ϕ, $\ker\phi$, is $\{g \in G : \phi(g) = e_H\}$.

Proposition 1.34

If $\phi : G \to H$ is a homomorphism, then

1. $\operatorname{im}\phi \leq H$;
2. $\ker\phi \leq G$.

Proof.

1. $e_H = \phi(e_G) \in \operatorname{im}\phi$, so we have the identity. $\phi(g_1)\phi(g_2) = \phi(g_1g_2) \in \operatorname{im}\phi$, so we have closure. Finally, by lemma 1.30, $\phi(g)^{-1} = \phi(g^{-1}) \in \operatorname{im}\phi$, so we have inverse.
2. $\phi(e_G) = e_H$, so $e_G \in \ker\phi$, so we have the identity. For $g_1, g_2 \in \ker\phi$, $\phi(g_1g_2) = \phi(g_1)\phi(g_2) = e_H \cdot e_H = e_H$, so $g_1g_2 \in \ker\phi$, so we have closure. Finally, by lemma 1.30, $\phi(g^{-1}) = \phi(g)^{-1} = e_H^{-1} = e_H$, so we have inverse.

■

Proposition 1.35

Let $\phi : G \to H$ be a homomorphism.

1. ϕ is *surjective* if and only if $\operatorname{im}\phi = H$;
2. ϕ is *injective* if and only if $\ker\phi = \{e_G\}$.

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Proof. The surjectivity statement is very clear. For the injectivity, we will show each direction as follows:

$\Longrightarrow$ If ϕ is injective, then there is only one element that maps to e_H, i.e. the kernel must have one element. That element clearly must be e_G.

$\Longleftarrow$ Suppose $\ker\phi = \{e_G\}$, and $\phi(g_1) = \phi(g_2)$. Then $\phi(g_1 g_2^{-1}) = \phi(g_1)\phi(g_2)^{-1} = e_H$. Hence, $g_1 g_2^{-1} \in \ker\phi \implies g_1 g_2^{-1} = e_G \implies g_1 = g_2$, which was what we wanted.

■

Combining the two parts of the proposition, we see that a homomorphism ϕ is an isomorphism iff and only if $\operatorname{im}\phi = H$ and $\ker\phi = \{e_G\}$.

§1.5 Cyclic groups

Definition 1.36 (Cyclic groups)**.** A group G is *cyclic* if there exists $g \in G$ such that $G = \{g^k : k \in \mathbb{Z}\} = \langle g\rangle$. g is called a *generator* of G.

Example

Let us have some examples.

- $C_n = \{z \in \mathbb{C} : z^n = 1\}$ is cyclic, with generator $e^{\frac{2\pi i}{n}}$;
- $\mathbb{Z}$ is cyclic, with generator 1;
- $\mathbb{Z}_n$ is cyclic, since $\mathbb{Z}_n \cong C_n$.

Theorem 1.37

If G is cyclic, then either

- $G \cong C_n$ for some n;
- or $G \cong \mathbb{Z}$.

Proof. Let G be a cyclic group with generator g. Let

$$S = \{k \in \mathbb{N} : g^k = e\}$$

and let

$$n = \begin{cases} \min S & S \neq \emptyset \\ \infty & \text{o/w} \end{cases}$$

We now split into cases.

Case 1. $n = \infty$. Define

$$\begin{aligned} \phi : \mathbb{Z} &\to G \\ k &\mapsto g^k \end{aligned}$$

which we would like to be an isomorphism. Since $\phi(k+l) = g^{k+l} = g^k g^l = \phi(k)\phi(l)$, ϕ is a homomorphism. Since G is cyclic, every element of G has the form g^k for some k, so ϕ is surjective. To prove injectivity, suppose for contradiction $0 \neq k \in \ker \phi$ – since $\ker \phi \leq \mathbb{Z}$, we may replace k by $-k$ if necessary, and assume $k > 0$. But then $k \in S$, which is a contradiction. The result follows.

Case 2. $n < \infty$. Define

$$\begin{aligned} \phi : \mathbb{Z}_n &\to G \\ k &\mapsto g^k \end{aligned}$$

as before. Since $k + l = qn + (k +_n l)$ for some $q \in \mathbb{Z}$,

$$\begin{aligned} \phi(k)\phi(l) = g^k g^l = g^{k+l} &= g^{qn+(k+_n l)} \\ &= g^{k+_n l} = \phi(k +_n l) \end{aligned}$$

since $g^n = e$. Hence ϕ is a homomorphism. Since G is cyclic, every element is g^k for some $k \in \mathbb{Z}$. Dividing k by n, we obtain

$$k = qn + r$$

for $q \in \mathbb{R}$ and $r \in \mathbb{Z}_n$. Hence,

$$g^k = g^{qn+r} = g^r = \phi(r)$$

as desired. To prove injectivity, suppose that $\phi(k) = e$ for some $k \in \mathbb{Z}_n$, that is $0 \leq k < n$. On the other hand, $k \in S$, so, if $k \neq 0$, $k \geq n$ by minimality of n, a contradiction. Hence $k = 0$, so $\ker \phi = \{0\}$ and ϕ is injective as required. Hence $G \cong \mathbb{Z}_n \cong C_n$ as required. ■

Because of the theorem, we will write C_n for any cyclic group of order n, and write $C_\infty \cong \mathbb{Z}$.

Definition 1.38 (Order of an element)**.** For any group G and $g \in G$, let

$$\langle g \rangle = \{g^k : k \in \mathbb{Z}\} \leq G$$

Since $\langle g \rangle$ is cyclic, $\langle g \rangle \cong C_n$ for some $n \in \mathbb{N} \cup \{\infty\}$. This is the *order* of g, $o(g)$ or $|g|$.

§1.6 Dihedral groups revisited

We would like to be able to tell if a group is dihedral just by checking some algebraic facts.

Lemma 1.39

Suppose x, $y \in G$ satisfy the dihedral relation $xy = yx^{-1}$. Then

$$yx^l = x^{-l}y$$

for any $l \in \mathbb{Z}$.

Proof. First, take $l > 0$, then

$$yx^{-l} = x^l y$$

by induction. Furthermore, if $y^2 = e$, then

$$yx^l = y(x^l y)y = y(yx^{-l})y = x^{-l}y$$

so that $yx^l = x^{-l}y$ for any $l \in \mathbb{Z}$. ∎

Lemma 1.40

Let G be a group, and a, $b \in G$ such that

- $a^n = e$ for some $n \geq 3$;
- $b^2 = e$;
- $ab = ba^{-1}$.

Then $\phi(r) = a$ and $\phi(s) = b$ defines a homomorphism $\phi : D_{2n} \to G$ sending r to a and s to b.

Proof. There are four cases to check.

1. $\phi(r^k)\phi(r^l) = a^k a^l = a^{k+l} = a^{k+_n l} = \phi(r^{k+_n l}) = \phi(r^{k+l})$;
2. $\phi(r^k)\phi(r^l s) = a^k a^l b = a^{k+l}b = a^{k+_n l}b = \phi(r^{k+_n l}s) = \phi(r^{k+l}s)$;
3. $\phi(r^k s)\phi(r^l) = a^k b a^l = a^{k-l}b = \phi(r^{k-l}s) = \phi(r^k s r^l)$ by lemma 1.39.
4. $\phi(r^k s)\phi(r^l s) = a^k b a^l b = a^{k-l}b^2 = a^{k-l} = \phi(r^{k-l}) = \phi(r^{k-l}s^2) = \phi(r^k s r^l s)$, also by lemma 1.39.

∎

Proposition 1.41

Suppose G has a generating set $\{a, b\}$ such that

- $a^n = e$ for some $n \geq 3$;
- $b^2 = e$;
- $ab = ba^{-1}$;
- $|G| = 2n$.

Then $G \cong D_{2n}$.

Proof. By the lemma, there is a homomorphism $\phi : D_{2n} \to G$. ϕ is surjective since $\{a, b\}$ generate G. Since $|D_{2n}| = 2n = |G|$, it must be a bijective. The result follows. ∎

Lecture 8 – 27/10/25

§2 Lagrange's theorem

This very important theorem helps us think about the orders of groups and their subgroups.

Theorem 2.1 (Lagrange)
If $H \leq G$ and $|G| < \infty$, $|H| \mid |G|$.

§2.1 Cosets

Definition 2.2. Left cosets Let $H \leq G$ and $g \in G$. The corresponding *left coset* is

$$gH = \{gh : h \in H\} \subseteq G$$

The set of all left cosets of H is denoted by G/H.

> **Remark.** We may equally define right cosets, Hg – the set of right cosets of H in G is denoted by $H \backslash G$.

Lemma 2.3
If $H \leq G$, the left cosets of H partition G – that is, every element of G is in exactly one coset. In other words, the following hold:

1. $$G = \bigcup_{gH \in G/H} gH$$
2. If $g_1H \cap g_2H \neq \emptyset$, then $g_1H = g_2H$.

Proof. 1. For any $g \in G$, $g = g \cdot e \in gH$, and so clearly $\bigcup gH = G$.

2. Suppose $g_1H \cap g_2H \neq \emptyset$, so there exists k with $k \in g_1H$, $k \in g_2H$, so $k = g_1h_1 = g_2h_2$ for some $h_1, h_2 \in H$. Thus $g_1 = g_2(h_2h_1^{-1}) \in g_2H$.
   Furthermore, for any $H \in H$, $g_1h = g_2((h_2h_1^{-1})h) \in g_2H$, so $g_1H \subseteq g_2H$. But the same reasoning would show $g_2H \subseteq g_1H$, and so we are done.

∎

DIAGRAM: G as a rectangle made of rectangles labelled with cosets.
But turns out things are even nicer than this – all the cosets have the same size.

Lemma 2.4

Let $H \leq G$ – then there is a bijection $H \to gH$ for any $g \in G$.

Proof. We claim the map $H \to gH$, $h \mapsto gh$ is a bijection. Observe we have an inverse $gh \to H$, $gh \mapsto g^{-1}(gh) = h$. This concludes the proof. ∎

Corollary 2.5

All left cosets have the same size.

DIAGRAM: G as a rectangle made of equally sized rectangles.

Definition 2.6 (Index of a subgroup)**.** The *index* of a subgroup $H \leq G$ is $|G : H| := |G/H|$, the number of cosets of H in G.

> **Remark.** One could show that all of these results continue to hold for right cosets.

Theorem 2.7 (Lagrange)

If $|G| < \infty$, then

$$|G| = |G : H||H|$$

for any $H \leq G$.

Proof. Since left cosets partition G, $|G|$ is the sum of the sizes of all the cosets in G, which is $|G : H||H|$. ∎

§2.2 Consequences of Lagrange's theorem

Corollary 2.8

If $|G| < \infty$ and $g \in G$, then $o(g) \mid |G|$.

Proof. $o(g) = |\langle g \rangle|$. ∎

Corollary 2.9

If $|G| < \infty$ and $g \in G$, then $g^{|G|} = e$.

Proof. Since $o(g) \mid |G|$, $g^{|G|} = (g^{o(g)})^{G/o(g)} = e^{G/o(g)} = e$. ∎

Corollary 2.10

If $|G|$ is prime, then G is cyclic.

Proof. Let $g \in G$, $g \neq e$. Then $o(g) | |G|$, so, since $|G|$ prime, $o(g) = |G|$ (since $o(g) \neq e$). Therefore, $\langle g \rangle = G$. ■

In fact, Lagrange's theorem implies an important result from number theory – the Fermat-Euler theorem.

Definition 2.11. The *Euler totient function* is given by

$$\varphi(n) = |\{x \in \mathbb{Z}_n : \gcd(x, n) = 1\}|$$

Lecture 9 – 29/10/25

Let $\times_n$ denote multiplication modulo n (on $\mathbb{Z}_n$). Recall that $x \in \mathbb{Z}_n$ has a multiplicative inverse modulo n if and only if $\gcd(x, n) = 1$. Thus,

$$\mathbb{Z}_n^\times = \{x \in \mathbb{Z}_n : \gcd(x, n) = 1\}$$

is an abelian group under multiplication.

Theorem 2.12 (Fermat-Euler)
Let $x, n \in \mathbb{Z}$ with $\gcd(x, n) = 1$. Then

$$x^{\varphi(n)} \equiv 1 \pmod{n}$$

Proof. By corollary 2.9, $x^{|\mathbb{Z}_n^\times|} = 1$ for any $x \in \mathbb{Z}_n^\times$. But $|\mathbb{Z}_n^\times| = \varphi(n)$, so $x^{\varphi(n)} = 1$ in $\mathbb{Z}_n^\times$, and so $x^{\varphi(n)} \equiv 1 \pmod{n}$ as required. ■

§3 Group actions

Groups become groups of symmetries when we make them act on symmetric objects.

Definition 3.1. An *action* of a group G on a set X is a map

$$\begin{aligned} G \times X &\to X \\ (g, x) &\mapsto gx \end{aligned}$$

such that

- $ex = x$ for all $x \in X$;
- $(gh)x = g(hx)$ for all $g, h \in G$, $x \in X$.

Example

- For any group G and a set X, $gx = x$ for all $g \in G$ defines the *trivial action*.
- $\mathrm{Sym}(X) \curvearrowright X$ for any set X by $fx = f(x)$ (remember f is a permutation).
- If $G \curvearrowright X$ and $H \leq G$, then $H \curvearrowright X$, so, for example, $\mathrm{Isom}(\mathbb{C}) \curvearrowright \mathbb{C}$.
- D_{2n} acts on X_n (the regular n-gon), whether 'filled in' or just the edges or vertices.
- Every group G acts on itself by the *left regular action* – just normal group multiplication on the left.

Theorem 3.2

An action of a group G on a set X is equivalent to a homomorphism

$$\phi : G \to \mathrm{Sym}(X)$$

Proof. Suppose $G \curvearrowright X$, and consider

$$\begin{aligned} t_g : X &\to X \\ x &\mapsto gx \end{aligned}$$

for any $g \in G$. Then $t_{g^{-1}} \circ t_g(x) = t_{g^{-1}}(gx) = g^{-1}(gx) = ex = x$, and so t_g has an inverse. Thus t_g is a bijection and in particular a permutation, so $t_g \in \mathrm{Sym}(X)$. Let us then define

$$\begin{aligned} \phi : G &\to \mathrm{Sym}(X) \\ g &\mapsto t_g \end{aligned}$$

We now wish to show that ϕ is a homomorphism. Observe

$$\begin{aligned} t_g \circ t_h(x) &= t_g(hx) \\ &= ghx \\ &= t_{gh}(x) \end{aligned}$$

This concludes the proof that every action corresponds to a homomorphism. Let us now prove the other direction. Given a homomorphism

$$\phi : G \to \mathrm{Sym}(X)$$

we define an action $G \curvearrowright X$ by $gx = \phi(g)(x)$. Let us now check that this is in fact an action:

- $ex = \phi(e)x = \mathrm{id}_X(x) = x$;
- $(gh)x = \phi(gh)(x) = \phi(g) \circ \phi(h)(x) = g(hx)$.

Thus we are done. ∎

Theorem 3.3 (Cayley)

Every group G is isomorphic to a subgroup of $\mathrm{Sym}(X)$ for some X. Furthermore, if $|G| < \infty$, we may require $|X| < \infty$.

Proof. Let $X = G$, and consider the left regular action $G \curvearrowright X = G$. By theorem 3.2, this is equivalent to a homomorphism

$$\phi : G \to \mathrm{Sym}(X)$$

Let $H = \operatorname{im}\phi \leq \mathrm{Sym}(X)$, so we may view ϕ as a homomorphism $G \to H$, and we claim this is in fact an isomorphism. It is surjective by the definition of H, so it remains to show it is injective. For this we will claim $\ker\phi = \{e\}$. Observe $g \in \ker\phi \implies \phi(g) = \mathrm{id}_X$, so $g\gamma = \gamma$ for all $\gamma \in G$. From this it is clear that $g = e$. Thus $G \cong H \leq \mathrm{Sym}(X)$, which was what we wanted. Since $G = X$, it is certainly true that $|G| < \infty \implies |X| < \infty$. ■

It turns out actions are extremely useful and we can use them to learn a great deal about groups.

Definition 3.4 (Orbits and stabilisers)**.** Let $G \curvearrowright X$ and $x \in X$.

- The *orbit* of x is the set $Gx = \{gx : g \in G\}$.
- The *stabiliser* of x is the set $\mathrm{Stab}_G(x) = G_x = \{g \in G : gx = x\}$.

If $Gx = X$ for any x, we say the action is *transitive*. If every element $g \neq e \in G$ has some $x \in X$ such that $gx \neq x$, then $G \curvearrowright X$ is called *faithful*.

> **Remark.** $G \curvearrowright X$ is faithful if the associated homomorphism is injective $\iff$ the kernel is trivial.

Lecture 10 – 31/10/25

Proposition 3.5

Suppose $G \curvearrowright X$, then

- $\mathrm{Stab}_G(x) \leq G$ for any $x \in X$;
- The orbits $\{Gy : y \in X\}$ form a partition of X.

Proof.
- We simply check that the stabiliser satisfies the definition of a subgroup.
  - *Closure under multiplication.* If $g, h \in \mathrm{Stab}_G(x)$, $(gh)x = g(hx) = gx = x$, so $gh \in \mathrm{Stab}_G(x)$;
  - *Closure under inversion.* If $g \in \mathrm{Stab}_G(x)$, $g^{-1}x = g^{-1}(gx) = (g^{-1}g)x = ex = x$, so $g^{-1} \in \mathrm{Stab}_G(x)$;
  - *Identity.* $ex = x$ by definition, so $e \in \mathrm{Stab}_G(x)$.

Hence $\mathrm{Stab}_G(x) \leq G$ as required.

- We must show every element is in an orbit, and if two orbits overlap, they are the same orbit.
  - Every element is in its own orbit – this shows the first part.
  - Now suppose $Gx_1 \cap Gx_2 \neq \emptyset$, so we have $y \in Gx_1$ and $\in Gx_2$, which means $y = g_1x_1 = g_2x_2$ for some $g_1, g_2 \in G$. Then

$$\begin{aligned} x_1 &= (g_1^{-1}g_1)x_1 \\ &= g_1^{-1}(g_1x_1) \\ &= g_1^{-1}(g_2x_2) \\ &= (g_1^{-1}g_2)x_2 \in Gx_2 \end{aligned}$$

    In fact, for any $gx_1 \in Gx_1$, $gx_1 = g(g_1^{-1}g_2)x_2 \in Gx_2$, so $Gx_1 \subseteq Gx_2$, and likewise $Gx_2 \subseteq Gx_1$, which concludes the proof.

∎

Example

Consider $D_{2n} \curvearrowright X_n$, the regular n-gon. Let us attempt to find the orbits and stabilisers associated with this action in the case $x = 1$

Clearly $r^j s(x) = r^j(x) = e^{2\pi ij/n}$, and so we can take x to any other element – that is, $D_{2n}x = X_n$. From this and proposition 3.5, it's clear that $D_{2n}x = X_n$ for any $x \in X_n$. A similar calculation makes clear that $\mathrm{Stab}_G(x) = \{e, s\}$.

Theorem 3.6 (Orbit-stabiliser theorem)

Suppose $G \curvearrowright X$, and let $x \in X$. The formula $g\,\mathrm{Stab}_G(x) \mapsto gx$ gives a well-defined bijection $G/\mathrm{Stab}_G(x) \to Gx$ – *cosets of the stabiliser are in bijection with orbits.*

Corollary 3.7

If $G \curvearrowright X$ and $x \in X$, then

$$|G| = |Gx||\mathrm{Stab}_G(x)|$$

Proof of corollary. The theorem gives that $|Gx| = |G/\mathrm{Stab}_G(x)|$, since they are in bijection. But $|G/\mathrm{Stab}_G(x)| = |G|/|\mathrm{Stab}_G(x)|$ by Lagrange, which concludes the proof. ∎

Example

Consider $D_{2n} \curvearrowright X_n$. We saw that, if $x \in X_n$, then $|D_{2n}| = n$ and $|\operatorname{Stab}_{D_{2n}}(x)| = 2$, so $|D_{2n}| = 2n$. This proof is circular, but this circularity could be resolved if we had approached things in a different way.

Proof of theorem. Let $S = \operatorname{Stab}_G(x)$, and let our as yet not well defined map Φ be such that $\Phi(gS) = gx$.

- *Well-definedness.* For any $g_1, g_2 \in G$ with $g_1S = g_2S$, we must have $\Phi(g_1S) = \Phi(g_2S)$, i.e. $g_1x = g_2x$. Since $g_1S = g_2S$, we have $s \in S$ with $g_1 = g_2s$. Then $g_1x = (g_2s)x = g_2(sx) = g_2x$ as required.
- *Surjectivity.* For any $gx \in Gx$, we clearly have $\Phi(gS) = gx$.
- *Injectivity.* Suppose $\Phi(g_1S) = \Phi(g_2S)$, that is $g_1x = g_2x$. Let $s = g_2^{-1}g_1$, so $sx = (g_2^{-1}g_1)x = x$, and so $s \in S$. But $g_1 = g_2s$, so $g_1 \in g_2S$, and so $g_1S = g_2S$ as required (since cosets partition G).

■

Example (Symmetries of a cube)

Let G be the group of isometries of a cube, and let x be the centre of a face. Clearly $|Gx| = 6$, since there are 6 other faces to take x to, and $\operatorname{Stab}_G(x)$ must fix the face on which x lies, so it is isomorphic D_8. Thus $|\operatorname{Stab}_G(x)| = 8$, and so $|G| = 48$. Nice!

Lecture 11 – 03/11/25

The next theorem is another cool application of the orbit-stabiliser theorem.

Theorem 3.8 (Cauchy)

If G is a finite group and p is a prime dividing $|G|$, there exists $g \in G$ with $o(g) = p$.

Proof. Consider the following set X consisting of p-tuples of elements of G:

$$X = \{(g_1, \ldots, g_p) : g_i \in G, g_1 \cdots g_p = e\}$$

Define an action of C_p on X: if $C_p = \{1, t, t^2, \ldots, t^{p-1}\}$, let

$$t^k(g_1, \ldots, g_p) = (g_{k+1}, \ldots, g_p, g_1, \ldots, g_k)$$

in other words, it cyclically permutes the tuple. Let us now verify that this is a bona fide action. It's easy to see that $t^k \cdot t^l(g_1, \ldots, g_p) = t^{k+l}(g_1, \ldots, g_p)$, and so it remains to verify

$$(g_{k+1}, \ldots, g_p, g_1, \ldots, g_k) \in X$$

Let us then suppose $g_1 \cdots g_p = e$. For convenience, write

$$a = g_1 \cdots g_k$$

$$b = g_{k+1} \cdots g_p$$

We know that $a \cdot b = e$, so $b = a^{-1}$ and hence $b \cdot a = e$ – a and b commute. From this it follows that

$$g_{k+1} \dots g_p \cdot g_1 \dots g_k = e$$

as required. Let us now compute $|X|$. Observe that, for any choices $g_1, \dots, g_{p-1} \in G$, $g_p = g_{p-1}^{-1} \cdots g_1^{-1}$ is the unique choice for g_n such that $(g_1, \dots, g_p) \in X$, and so $|X| = |G|^{p-1}$. Furthermore, the action of C_p partitions X into orbits:

$$X = C_p x_1 \cup C_p x_2 \cup \cdots \cup C_p x_k$$

Then, by corollary 3.7 (orbit-stabiliser), for each j,

$$|C_p x_j| \;\big|\; |C_p| = p$$

so either $|C_p x_j| = 1$ or $|C_p x_j| = p$. Let l be the number of orbits of size 1, so after renumbering we may assume

$$|C_p x_j| = \begin{cases} 1 & 1 \le j \le l \\ p & l < j \le k \end{cases}$$

Since the orbits partition X,

$$\begin{aligned} |G|^{p-1} &= |X| \\ &= \sum_{j=1}^{k} |C_p x_j| \\ &= l + p(k - l) \end{aligned}$$

and so, since $p \mid |G|$, $p \mid l$. But note that $x = (g_1, \dots, g_p)$ has an orbit of size 1 if and only if

$$g_1 = \cdots = g_p$$

and we can observe that $(e, \dots, e)$ is clearly in X and has an orbit of size 1, so $l > 0$. But then $l \ge p > 1$, so there must be a further trivial orbit

$$(g, \dots, g) \in X$$

with $g \neq e$ and $g^p = e$. Thus $o(g) \mid p$, and so $o(g) = p$ as required.

■

§3.1 Conjugation

Definition 3.9. Let G be a group and $g \in G$, For any $h \in G$,

$$hgh^{-1}$$

is the *conjugate* of g by h.

Intuition:

- hgh^{-1} has the 'same shape' as g;
- hgh^{-1} corresponds to 'changing the coordinates' of g.

Definition 3.10. The *conjugacy class* of $g \in G$ is

$$\mathrm{ccl}(g) = \{hgh^{-1} \mid g \in G\}$$

Example
- If G is abelian, then $hgh^{-1} = ghh^{-1} = g$, so the only conjugate of g is g itself.
- D_{2n} has one conjugacy class of reflections if n is odd and two if n is even.

G acts on itself by conjugation, where we define $g\gamma = g\gamma g^{-1}$. This is an important action which is *very* different to the regular action. Notably, $\mathrm{ccl}(g)$ is the orbit of g under this action.

Definition 3.11. The *centraliser* of $g \in G$ is defined as

$$C_G(g) = \{h \in G : hgh^{-1} = g \iff hg = gh\}$$

which is the stabiliser of g under the conjugation action. It is also clearly the set of elements that commute with g.

Definition 3.12. The *centre* of G is

$$\begin{aligned} Z(G) &= \{h \in G : hg = gh \text{ for all } g \in G\} \\ &= \bigcap_{g \in G} C_G(g) \end{aligned}$$

the set of elements that commute with all elements of G.

Lecture 12 – 05/11/25

§4 The Möbius group

This group is *almost* a group of bijections of $\mathbb{C}$, *but* we need to add a formal 'point at ∞'.

Definition 4.1. The *Riemann sphere* $\mathbb{C}_\infty$ is $\mathbb{C} \cup \{\infty\}$.

> **Remark.** Why a sphere? We see there is a map $\pi : \mathbb{C} \to$ a sphere S^2, called *stereographic projection*, which misses out the north pole. Hence, by adding ∞ to $\mathbb{C}$ and mapping this to the north pole, we have a bijection between $\mathbb{C}_\infty$ and S^2. DIAGRAM!

Definition 4.2. Let a, b, c, $d \in \mathbb{C}$, with $ad - bc \neq 0$. If $c \neq 0$, then the corresponding *Möbius transformation*

$$\begin{aligned} \mu : \mathbb{C}_\infty &\to \mathbb{C}_\infty \\ z &\mapsto \begin{cases} \dfrac{az+b}{cz+d} & z \neq -\dfrac{d}{c} \\ \infty & z = -\dfrac{d}{c} \\ \dfrac{a}{c} & z = \infty \end{cases} \end{aligned}$$

If $c = 0$, then

$$\mu : \mathbb{C}_\infty \to \mathbb{C}_\infty$$

$$z \mapsto \begin{cases} \dfrac{az+b}{d} & z \neq -\dfrac{d}{c} \\ \infty & z = \infty \end{cases}$$

But we usually just write

$$\mu(z) = \frac{az+b}{cz+d}$$

and all the edge cases are implicit.

The *Möbius* group $\mathcal{M}$ is then the group of Möbius transformations under composition.

Proposition 4.3

If

$$\mu_1(z) = \frac{a_1 z + b_1}{c_1 z + d_1}, \quad \mu_2(z) = \frac{a_2 z + b_2}{c_2 z + d_2}$$

then

$$\mu_1 \circ \mu_2(z) = \frac{(a_1 a_2 + b_1 c_2)z + (a_1 b_2 + b_1 d_2)}{(c_1 a_2 + d_1 c_2)z + (c_1 b_2 + d_1 d_2)}$$

Proof. Exercise. ■

Remark. Compare

$$\begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}\begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} = \begin{pmatrix} a_1 a_2 + b_1 c_2 & a_1 b_2 + b_1 d_2 \\ c_1 a_2 + d_1 c_2 & c_1 b_2 + d_1 d_2 \end{pmatrix}$$

so the coefficients in the composition are given by matrix multiplication.

Theorem 4.4

$(\mathcal{M}, \circ, \mathrm{id})$ is a group.

Proof. Composition of functions is associative, so that's ok. The identity is given by

$$\mathrm{id} : z \mapsto z = \frac{1 \cdot z + 0}{0 \cdot z + 1}$$

so that's ok too. Now let's think about inverses. For a Möbius transformation

$$\mu : z \mapsto \frac{az+b}{cz+d}$$

let

$$\nu : z \mapsto \frac{dz-b}{-cz+a}$$

(whose coefficients are given by the matrix inverse of those of μ). It follows from proposition 4.3 that $\mu \circ \nu = \mathrm{id}$ (here we are using $ad - bc \neq 0$). Closure again follows from proposition 4.3 and the comparison with matrix multiplication. ■

One important way to study Möbius transformations is via their fixed points.

Definition 4.5. Suppose $f : X \to X$ is a permutation. If $f(x) = x$ for some $x \in X$, then x is called a *fixed point* of f.

Lemma 4.6 (3-point lemma for Möbius transformations)

If $\mu \in \mathcal{M}$ fixes three distinct points w_1, w_2, $w_3 \in \mathbb{C}_\infty$, then $\mu = \mathrm{id}$.

Proof. Let
$$\mu(z) = \frac{az+b}{cz+d}$$
Then, a fixed point w_i satisfies
$$w_i = \frac{aw_i+b}{cw_i+d}$$
If $w_1 = \infty$, then $c = 0$, so w_2 and w_3 satisfy
$$\begin{aligned} w_i &= \frac{aw_i+b}{d} \\ \iff (a-d)w_i + b &= 0 \end{aligned}$$
which has two distinct roots w_2 and w_3. It follows that $a = d$, $b = 0$. Then it's clear that $\mu = \mathrm{id}$.

Now suppose that none of the w_i are infinity. Again,
$$\begin{aligned} w_i &= \frac{aw_i+b}{cw_i+d} \\ \implies cw_i^2 + dw_i &= aw_i + b \\ \implies cw_i^2 + (d-a)w_i - b &= 0 \end{aligned}$$
Now we have a quadratic with 3 distinct roots, so again, we must have $b = c = 0$, $a = d$. As before, $\mu = \mathrm{id}$. ■

Proposition 4.7

Every $\mu \in \mathcal{M}$ has at least one fixed point.

Sketch proof. As above – every quadratic has at least one root in $\mathbb{C}$. Full proof an exercise ■

Example
- $\mu(z) = z + 1$ has ∞ as its unique fixed point – $\mathrm{Fix}(\mu) = \{\infty\}$.
- $\mu(z) = 2z$ has 0 and ∞ as its two fixed points – $\mathrm{Fix}(\mu) = \{0, \infty\}$.

Lemma 4.8 (Triple transitivity)

For any two triples of distinct points z_1, z_2, z_3 and w_1, w_2, $w_3 \in \mathbb{C}_\infty$, there exists $\mu \in \mathcal{M}$ such that $\mu(z_i) = w_i$ for each i.

Proof. Let
$$\alpha(z) = \frac{z_2 - z_3}{z_2 - z_1} \cdot \frac{z - z_1}{z - z_3}$$
(modify appropriately if any $z_i = \infty$). Note
$$\begin{aligned} z_1 &\longmapsto 0 \\ z_2 &\longmapsto \infty \\ z_3 &\longmapsto 1 \end{aligned}$$
Similarly, we can write down β which sends w_1, w_2, w_3 to these three points. Then, $\beta^{-1} \circ \alpha$ is the Möbius transformation we're after.

■

> **Remark.** Named for the fact that it's a strengthening of transitivity when interpreting $\mathcal{M}$ as acting on $\mathbb{C}_\infty$.

> **Remark.** The three-point lemma implies that this μ is unique. We then say further that the action $\mathcal{M} \curvearrowright \mathbb{C}_\infty$ is *sharply triply transitive.*

Definition 4.9. Let z_1, z_2, z_3, $z_4 \in \mathbb{C}_\infty$ be distinct. Since $\mathcal{M} \curvearrowright \mathbb{C}_\infty$ sharply triply transitively, there is a unique α such that $\alpha(z_1) = 0$, $\alpha(z_2) = 1$, and $\alpha(z_3) = \infty$. Then the *cross ratio*
$$[z_1, z_2, z_3, z_4] = \alpha(z_4)$$
If we really want, we can find an explicit formula using our explicit formula for α:
$$[z_1, z_2, z_3, z_4] = \frac{z_2 - z_3}{z_2 - z_1} \cdot \frac{z_4 - z_1}{z_4 - z_3}$$

Lecture 13 – 07/11/25

Let's try and find a generating set for the Möbius group.

Proposition 4.10 (Generating set for the Möbius group)
$\mathcal{M}$ is generated by the set of elements of the following 3 forms:

- $\alpha_a : z \mapsto az$, $a \neq 0$;
- $\beta_b : z \mapsto z + b$, $b \in \mathbb{C}$;
- $\gamma : z \to \frac{1}{z}$.

Proof. Let $\mu \in \mathcal{M}$ be arbitrary, and let $z_1 = \mu(0)$, $z_2 = \mu(1)$, $z_3 = \mu(\infty)$.

1. We want to construct μ_1 such that $\mu_1(z_3) = \infty$. Either $z_3 = \infty$ and $\mu_1 = \text{id}$, or
$$\mu_1(z) = \frac{1}{z + b} = \gamma \circ \beta_b(z)$$
where $b = -z_3$. Now, let $z_1' = \mu_1(z_1)$, $z_2' = \mu_1(z_2)$.

2. Let $b' = -z_1'$ and $\mu_2 = \beta_b'$, that is $\mu_2(z) = z + b'$. Observe $\mu_2(\infty) = \infty$ and $\mu_2(z_1') = 0$. Then by constructikn we have

$$\begin{aligned} \mu_2 \circ \mu_1(z_3) &= \infty \\ \mu_2 \circ \mu_1(z_1) &= \mu_2(z_1') = 0 \end{aligned}$$

Lastly, let $z_2'' = \mu_2 \circ \mu_1(z_2) \neq 0, \infty$.

3. Let $a = \frac{1}{z_2''}$ and $\mu_3(z) = \alpha_a(z)$, that is $\frac{z}{z_2''}$.

It's now clear that 0, 1 and ∞ have the same image under μ and $\mu_1^{-1} \circ \mu_2^{-1} \circ \mu_3^{-1}$, from which it follows by the three-point lemma that $\mu = \mu_1^{-1} \circ \mu_2^{-1} \circ \mu_3^{-1}$. Hence our generating set generates μ as required. ■

Definition 4.11. A *circle* in $\mathbb{C}_\infty$ is

- a standard Euclidean circle in $\mathbb{C}$;
- or $l \cup \{\infty\}$ where $l \subseteq \mathbb{C}$ is a Euclidean straight line.

> **Remark 4.12.** For completeness, let us add that Euclidean circles in $\mathbb{C}$ have the form $|z - c| = r$, $c \in \mathbb{C}$, $r > 0 \in \mathbb{R}$, while Euclidean lines have the form $|z - a| = |z - b|$ for a, b not unique in $\mathbb{C}$.

Theorem 4.13 (Circles and Möbius transformations)
If $C \subseteq \mathbb{C}_\infty$ is a circle and $\mu \in \mathcal{M}$ is a Möbius transformation, then $\mu(C)$ is also a circle.

Proof. This is a great opportunity to apply proposition 4.10 – it is sufficient to check the result for only $\alpha_a(z) = az$, $\beta_b(z) = z + b$, $\gamma(z) = \frac{1}{z}$. Furthermore, it is really quite obvious that α_a and β_b preserve circles.

If C is a Euclidean circle, $\gamma(C)$ has equation

$$\begin{aligned} \left| \frac{1}{z} - c \right| &= r \\ \iff \left(\frac{1}{z} - c \right) \left(\frac{1}{\bar{z}} - \bar{c} \right) &= r^2 \\ \iff \frac{1}{|z|^2} - \frac{c}{z} - \frac{\bar{c}}{z} + |c|^2 - r^2 &= 0 \\ \iff (|c|^2 - r^2)|z|^2 - cz - \bar{c}\bar{z} + 1 &= 0 \end{aligned}$$

In the case $|c|^2 = r^2$, we have

$$\begin{aligned} cz + \bar{c}\bar{z} &= 1 \\ \iff \frac{z}{\bar{c}} + \frac{\bar{z}}{c} &= \frac{1}{|c|^2} \\ \iff |z|^2 &= |z|^2 - \frac{z}{\bar{c}} - \frac{\bar{z}}{c} + \frac{1}{|c|^2} \end{aligned}$$

$$= \left| z - \frac{1}{c} \right|^2$$

$$\iff |z| = \left| z - \frac{1}{c} \right|$$

which is a straight line – furthermore, it is clear 0 is on the circle, and this point will go to infinity. Otherwise, $|c|^2 \neq r^2$, so we have

$$\begin{aligned} |z|^2 - \left(\frac{c}{|c|^2 - r^2} \right) z - \left(\frac{\bar{c}}{|c|^2 - r^2} \right) \bar{z} + \frac{1}{|c|^2 - r^2} = 0 \\ \iff \left| z - \frac{\bar{c}}{|c|^2 - r^2} \right|^2 &= \frac{|c|^2}{(|c|^2 - r^2)^2} - \frac{1}{|c|^2 - r^2} \\ &= \frac{r^2}{(|c|^2 - r^2)^2} \end{aligned}$$

which is a circle. The case where C is a line union infinity is left as an exercise. ■

Corollary 4.14

Four points z_1, z_2, z_3, $z_4 \in \mathbb{C}$ lie on a circle if and only if

$$[z_1, z_2, z_3, z_4] \in \mathbb{R} \cup \{\infty\}$$

Proof. Let $\alpha \in \mathcal{M}$ be such that $\alpha(z_1) = 0$, $\alpha(z_2) = 1$, $\alpha(z_3) = \infty$, and $\alpha(z_4) = [z_1, z_2, z_3, z_4]$.

$\implies$ Suppose $z_1, \dots, z_4 \in C$ a circle. Then 0, 1, ∞ and $\alpha(z_4)$ will lie on $\alpha(C)$, also a circle. Since ∞ is on this circle, it is in fact a line union infinity, and so, since it passes through 0 and 1, it is $\mathbb{R} \cup \{\infty\}$, so $\alpha(z_4) \in \mathbb{R} \cup \{\infty\}$ as required.

$\impliedby$ Suppose $\alpha(z_4) \in \mathbb{R} \cup \{\infty\}$, and of course $\alpha(z_1) = 0$, $\alpha(z_2) = 1$, $\alpha(z_3) = \infty$ – so they all lie on a circle. Hence $z_1, \dots, z_4 \in \alpha^{-1}(\mathbb{R} \cup \{\infty\})$ which is a circle by the theorem.

■

§5 Finite groups

Let's categorise all the finite groups!

Lecture 14 – 10/11/25

- *Order 1*. If $|G| = 1$, then $G \cong \{e\}$, the trivial group.
- *Order 2*. If $|G| = 2$, since 2 is prime, $G \cong C_2$ by corollary 2.10.
- *Order 3*. As above, $G \cong C_3$.
- *Order 4*. If $|G| = 4$, we have C_4. But is that all?

Definition 5.1 (Direct product)**.** If G and H are groups, the *direct product* is

$$G \times H = \{(g, h) : g \in G, h \in H\}$$

with group operation

$$(g_1, h_1) \cdot (g_2, h_2) = (g_1 g_2, h_1 h_2)$$

and identity (e_G, e_H). Inverses are also pointwise – $(g, h)^{-1} = (g^{-1}, h^{-1})$.

Example (Klein 4-group)

The group

$$K_4 := C_2 \times C_2$$

is often called the Klein 4-group (it also makes sense to think of D_4 as K_4). Note that every element in K_4 has order 2, while there exist elements of order 4 in C_4, so $K_4 \ncong C_4$.

Theorem 5.2 (Direct product theorem)

If $H_1, H_2 \leq G$ and

- $H_1 \cap H_2 = \{e\}$;
- $h_1 h_2 = h_2 h_1$ for all $h_1 \in H_1$, $h_2 \in H_2$;
- $G = H_1 H_2$ – for all $g \in G$, $g = h_1 h_2$ for some $h_1 \in H_1$, $h_2 \in H_2$.

Then $G \cong H_1 \times H_2$.

Proof. Define

$$\begin{aligned} \Phi : H_1 \times H_2 &\to G \\ (h_1, h_2) &\mapsto h_1 h_2 \end{aligned}$$

We claim Φ is an isomorphism.

- Φ *is a homomorphism*. For any $h_1, h_1' \in H_1$, $h_2, h_2' \in H_2$, we have

$$\begin{aligned} \Phi(h_1, h_2)\Phi(h_1', h_2') &= h_1 h_2 h_1' h_2' \\ &= h_1 h_1' h_2 h_2' \\ &= \Phi(h_1 h_1', h_2 h_2') \\ &= \Phi((h_1, h_2)(h_1', h_2')) \end{aligned}$$

- Φ *is surjective*. Follows immediately from $G = H_1 H_2$.
- Φ *is injective*. We will show $\ker \Phi$ is trivial. Suppose $\Phi(h_1, h_2) = e$ for some $(h_1, h_2) \in H_1 \times H_2$. Then

$$\begin{aligned} & h_1 h_2 = e \\ \implies & H_1 \ni h_1 = h_2^{-1} \in H_2 \end{aligned}$$

so $h_1 = h_2^{-1} \in H_1 \cap H_2$. But $H_1 \cap H_2 = \{e\}$, so $h_1 = h_2 = e$. Thus (h_1, h_2) is the identity in $H_1 \times H_2$ as required.

■

> **Remark.** If we know $H_1 \cap H_2 = \{e\}$, then $|H_1 \times H_2| = |H_1||H_2|$. So if $|H_1||H_2| = |G|$, then we get that $G = H_1 H_2$ automatically.

Lemma 5.3 (Groups of order 4)
If $|G| = 4$, then either $G \cong C_4$ or $G \cong K_4$.

Proof. By Lagrange, every non-trivial element has order 2 or 4. If there is an element $g \in G$ of order 4, then $G = \langle g \rangle \cong C_4$. Otherwise, every non-trivial element of G has order 2. Let a, b be distinct non-trivial elements of G, and let $H_1 = \langle a \rangle$, $H_2 = \langle b \rangle$. Clearly $H_1 \cap H_2 = \{e\}$ and, since $|H_1| = |H_2| = 2$, we have $G = H_1 H_2$. Finally, since $o(ab) = 2$, $abab = e \implies ab = b^{-1}a^{-1} = ba$, so we have commutativity. Thus $G = H_1 \times H_2 \cong C_2 \times C_2$. ■

Theorem 5.4 (Chinese remainder theorem)
If $\gcd(m, n) = 1$, then $C_m \times C_n \cong C_{mn}$.

Proof. Let $C_{mn} = \langle g \rangle$. Let $H_1 = \langle g^n \rangle \cong C_m$, $H_2 = \langle g^m \rangle \cong C_n$. We will check the hypotheses of the direct product theorem.

- Observe $g^k \in H_1 \iff k = np + mnq$ for some $p, q \in \mathbb{Z}$, i.e. if and only if $n \mid k$. Similarly, $g^k \in H_2$ if and only if $m \mid k$. Hence $g^k \in H_1 \cap H_2$ if and only if $n, m \mid k \iff \operatorname{lcm}(m, n) \mid k$. But $\gcd(m, n) = 1 \implies \operatorname{lcm}(m, n) = mn$, so $mn \mid k$ – but $g^{mn} = e$, so $g^k = e$ as required.
- C_{mn} is abelian, so we definitely have commutativity.
- Immediate by the remark.

■

Let's continue classifying groups!

- *Order 5*. C_5 is of course the only possibility.
- *Order 6*.

Lemma 5.5 (Groups of order 6)
If $|G| = 6$, then $G \cong C_6$ or $G \cong D_6$.

Proof. By Cauchy's theorem, G has an element r of order 3 and an element s of order 2. Since $|\langle r \rangle| = 3$, $|G : \langle r \rangle| = 2$, and $s \notin \langle r \rangle$, so $s\langle r \rangle = G \setminus \langle r \rangle = \langle r \rangle s$. We conclude that $sr = r^i s$ for some $i = 0, 1, 2$.

Case 1. $i = 0$. Then $sr = s \implies r = e$, but this contradicts $o(r) = 3$.

Case 2. $i = 1$. Then $sr = rs$, from which it's clear that $\langle r \rangle$ and $\langle s \rangle$ commute. Then theorem 5.2 (the direct product theorem) applies, so $G \cong C_2 \times C_3 = C_6$ by theorem 5.4.

Case 3. Then we have $sr = r^2 s = r^{-1} s$ – this is the dihedral relation. We conclude $G \cong D_6$ by proposition 1.31 (to be added). ■

> **Remark.** S_3 is a non-abelian group of order 6, so this gives us a way to conclude $S_3 \cong D_6$.

Lecture 15 – 12/11/25

- *Order 7.* C_7 only.
- *Order 8.* Interesting! We have $C_2 \times C_2 \times C_2$, $C_4 \times C_2$, C_8 of course. We also have D_8. But there's another one!

Example 5.6 (The quaternion group)

Let

$$Q_8 = \left\{ \begin{pmatrix} \pm 1 & 0 \\ 0 & \pm 1 \end{pmatrix}, \begin{pmatrix} \pm i & 0 \\ 0 & \mp i \end{pmatrix}, \begin{pmatrix} 0 & \pm 1 \\ \mp 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & \pm i \\ \pm i & 0 \end{pmatrix} \right\}$$

One can check that this is a group. We usually use Hamilton's notation, where

$$1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \ i = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \ j = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \ k = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

so that the group consists of $\{\pm 1, \pm i, \pm j, \pm k\}$. Then we have various properties:

- $i^2 = j^2 = k^2 = -1$;
- $(-1)i = -i$, etc;
- $ij = k$, $jk = i$, $ki = j$;
- $ji = -k$, $kj = -i$, $ik = -j$.
- -1 commutes with everything.

But is this in our list? It's not abelian, so if it were, it would have to be D_8. But D_8 has too many elements of order 2.

Lemma 5.7 (Groups of order 8)

If $|G| = 8$, then $G \cong C_2 \times C_2 \times C_2$, $C_4 \times C_2$, C_8, D_8 or Q_8.

Proof. By Lagrange, every element has order 1, 2, 4 or 8.

- If we have an element of order 8, we're done ($G \cong C_8$).

- Suppose that every non-trivial element has order 2. We know it's abelian, and then by picking a, $b \neq a$, $c \neq a, b, ab$, we see that

  $$G = \langle a \rangle \times \langle b \rangle \times \langle c \rangle \cong C_2 \times C_2 \times C_2$$

  by applying the direct product theorem twice.

- Now we may suppose we have $a \in G$ with $o(a) = 4$, so $|G : \langle a \rangle| = 2$ – also let $b \in G \setminus \langle a \rangle$. Then

  $$b \langle a \rangle = G \setminus \langle a \rangle = \langle a \rangle b$$

  so that $ba = a^i b$ for some $i = 0, 1, 2, 3$.

  - $i = 0$, so $a = e$, which is a contradiction.
  - $i = 1$, so $ba = ab$, and it follows $ba^j = a^j b$ for all j. It follows from here that the whole group is abelian.
  - $i = 2$, so $ba = a^2 b \implies bab^{-1} = a^2$. But bab^{-1} is a conjugated by b, so must have the same order as a, i.e. 4, while a^2 clearly has order 2. So this is a contradiction.
  - $i = 3$, so $ba = a^3 b = a^{-1} b$.

  Let's take a break and think about powers of b. If $b^2 = ba^j$, then $b = a^j$, so $b \in \langle a \rangle$, but we picked $b \in G \setminus \langle a \rangle$. So $b^2 \in \langle a \rangle$, and we get four more cases.

  - $b^2 = e$.
  - $b^2 = a$. Then, since a has order 4, b has order 8, so we get C_8 again.
  - $b^2 = a^2$.
  - $b^2 = a^3 = a^{-1}$ – again, a^{-1} has order 4, and so we get C_8 once more.

  This leaves us with four combinations of cases to analyse.

  - G is abelian, $b^2 = e$. Then $\langle b \rangle \cong C_2$, $\langle a \rangle \cong C_4$, $\langle b \rangle \cap \langle a \rangle = \{e\}$, so $G \cong C_4 \times C_2$ by the direct product theorem.
  - G is abelian, $b^2 = a^2$. Then ab^{-1} has order 2, and it turns out that once again $G \cong C_4 \times C_2$ by the direct product theorem.
  - $ba = a^{-1} b$, $b^2 = e$, and we already knew $a^4 = e$. That's enough to show that $G \cong D_8$ with $r = a$ and $s = b$.
  - $ba = a^{-1} b$, $b^2 = a^2$. We claim $G \cong Q_8$. Let $i = a$, $j = b$, $k = ab$, $-1 = a^2 = b^2$. Then

    $$\begin{aligned} G &= \{e, a, a^2, a^3, b, ab, a^2 b, a^3 b\} \\ &= \{1, i, -1, -i, j, k, -j, -k\} \end{aligned}$$

    and we can check multiplication works. So $G \cong Q_8$ as required.

■

We have now categorised all the groups of order less than or equal to 8: $\{e\}$, C_2, C_3, C_4, $C_2 \times C_2$, C_5, C_6, D_6, C_7, C_8, $C_4 \times C_2$, $C_2 \times C_2 \times C_2$, D_8, Q_8.

§6 Quotients

§6.1 Normal subgroups

Definition 6.1. $H \leq G$ is called *normal* if $ghg^{-1} \in H$ for every $h \in H$ and $g \in G$. We write $H \lhd G$.

Example 6.2
- $\{e\} \lhd G$, $G \lhd G$ for all G;
- If G is abelian, $H \lhd G$ for all $H \leq G$;
- $\langle r \rangle \lhd D_{2n}$, but $\langle s \rangle$ is not a normal subgroup of D_{2n}.

Proposition 6.3
Let $\phi : G \to H$ be a homomorphism. Then $\ker \phi \lhd G$.

Proof. We have

$$\begin{aligned}\phi(ghg^{-1}) &= \phi(g)\phi(h)\phi(g)^{-1} \\ &= \phi(g)\phi(g)^{-1} \\ &= e\end{aligned}$$

so $ghg^{-1} \in \ker \phi$. Hence $\ker \phi \lhd G$ as required. ■

Lecture 16 – 14/11/25

Lemma 6.4
Suppose $H \leq G$. Then $H \lhd G$ if and only if $gH = Hg$ for all $g \in G$.

Proof. $\implies$ Let $h \in H$, $g \in G$. Since $H \lhd G$, $ghg^{-1} \in H$. Hence $gh = (ghg^{-1})g \in Hg$, so $gH \subseteq Hg$. Similarly, $Hg \subseteq gH$, so $gH = Hg$ as required.

$\impliedby$ Suppose $gH = Hg$. Let $h \in H$, so $gh \in gH = Hg$, so we have $h' \in H$ with $gh = h'g$. Hence $ghg^{-1} = h' \in H$ as required.

■

Theorem 6.5 (Quotient groups)
If $H \lhd G$, the set of left cosets G/H is a group with operation $(g_1H)(g_2H) = (g_1g_2)H$.

Proof. We need to check that the operation is well-defined and satisfies the group axioms.

- *Well-defined*. Suppose $g_1H = g_1'H$ and $g_2H = g_2'H$, so $Hg_2 = Hg_2'$ (since H normal in G). Hence, we have $h_1, h_2 \in H$ with $g_1 = g_1'h_1$ and $g_2 = h_2g_2'$. Therefore,

$$\begin{aligned} g_1g_2 &= (g_1'h_1)(h_2g_2') \\ &= g_1'(h_1h_2)g_2' \\ &= g_1'g_2'h_3 \end{aligned}$$

  for some $h_3 \in H$ by lemma 6.4. Therefore, $g_1g_2 \in g_1'g_2'H$, so $g_1g_2H = g_1'g_2'H$.

- *Associativity*. Obvious from the definition.

- *Identity*. H.

- *Inverses*. Clearly $g^{-1}H = (gH)^{-1}$.

■

Definition 6.6. If $H \triangleleft G$, then G/H, which is a group by theorem 6.5, is called the *quotient of G by H*.

Example 6.7

- $G/\{e\} \cong G$, $G/G \cong \{e\}$.
- Since $\mathbb{Z}$ is abelian, $n\mathbb{Z} \triangleleft \mathbb{Z}$ for any n, and $\mathbb{Z}/n\mathbb{Z}$ is cyclic of order n, with generator $1 + n\mathbb{Z}$, so $\mathbb{Z}/n\mathbb{Z} \cong C_n$.
- Let G be a group, $H \leq G$, and suppose $|G : H| = 2$. Then $H \triangleleft G$ as we've argued before. For example, $C_n \cong \langle r \rangle \triangleleft D_{2n}$, and $D_{2n}/C_n \cong C_2$.

Remark. Quotienting and the direct product are not opposites! $A/B \cong C \not\Rightarrow A \cong B \times C$.

Theorem 6.8 (Isomorphism theorem)

If $\phi : G \to H$ is a homomorphism, then

$$G/\ker\phi \cong \operatorname{im}\phi$$

Proof. Since $\ker\phi \triangleleft G$, the quotient $G/\ker\phi$ is indeed a group. Let's define

$$\begin{aligned} \bar{\phi} : G/\ker\phi &\to \operatorname{im}\phi \\ g\ker\phi &\mapsto \phi(g) \end{aligned}$$

We must perform all the usual checks.

- *Well-defined*. Suppose $g_1\ker\phi = g_2\ker\phi$, so $g_1 = g_2k$ for some $k \in \ker\phi$. Then

$$\begin{aligned} \bar{\phi}(g_1\ker\phi) &= \phi(g_1) \\ &= \phi(g_2k) \\ &= \phi(g_2)\phi(k) \\ &= \phi(g_2) \\ &= \bar{\phi}(g_2\ker\phi) \end{aligned}$$

- *Homomorphism*. For $g_1,\, g_2 \in G$,

$$\begin{aligned}\bar{\phi}((g_1 \ker\phi)(g_2 \ker\phi)) &= \bar{\phi}(g_1 g_2 \ker\phi)\\ &= \phi(g_1 g_2)\\ &= \phi(g_1)\phi(g_2)\\ &= \bar{\phi}(g_1 \ker\phi)\bar{\phi}(g_2 \ker\phi)\end{aligned}$$

- *Injective*. If $\bar{\phi}(g\ker\phi) = e$, then $\phi(g) = \bar{\phi}(g\ker\phi) = e$, so $g \in \ker\phi$. Hence $g\ker\phi = \ker\phi$, and so $\ker\bar{\phi} = \{\ker\phi\}$ is trivial, hence $\bar{\phi}$ is injective as required.

- *Surjective*. An element of $\operatorname{im}\phi$ looks like $\phi(g)$, which is $\bar{\phi}(g\ker\phi)$ as required.

$\blacksquare$

Example 6.9
- Since $\phi : \mathbb{Z} \to \mathbb{C}_\times$, $k \mapsto e^{2\pi i k/n}$ is a homomorphism with image C_n and kernel $n\mathbb{Z}$, we get that $\mathbb{Z}/n\mathbb{Z} \cong C_n$ by the isomorphism theorem.
- Similarly, $\phi : \mathbb{R} \to \mathbb{C}_\times$, $k \mapsto e^{2\pi i t}$ is a homomorphism with image $\{z : |z| = 1\} = U(1)$ and kernel $\mathbb{Z}$, we have $\mathbb{R}/\mathbb{Z} \cong U(1)$.

Lecture 17 – 17/11/25

Definition 6.10. A group G is *simple* if the only normal subgroups are $\{e\}$ and G itself. Thus every homomorphism out of G is either injective or trivial by the isomorphism theorem.

Example 6.11

C_p is simple for p a prime.

> **Remark.** The classification of finite simple groups is one of the great mathematical achievements of the 20th century.

§7 Permutations

Recall that a permutation of a set X is a bijection $X \to X$, and that $\operatorname{Sym}(X)$ is the group of all such permutations under composition.

Definition 7.1. Any list of k distinct elements $a_1, \ldots, a_k \in \{1, \ldots, n\}$ determines a *k-cycle*:

$$\sigma = (a_1 \; \ldots \; a_k)$$

which is a permutation defined as follows:

$$b \mapsto \begin{cases} a_{i+1} & b = a_i, i \neq k \\ a_1 & b = a_k \\ b & \text{otherwise} \end{cases}$$

Example 7.2

The permutation σ taking 1 to 3, 2 to 1 and 3 to 2 is a 3-cycle represented as

$$\begin{aligned} \sigma &= (1\ 3\ 2) \\ &= (3\ 2\ 1) \\ &= (2\ 1\ 3) \end{aligned}$$

The permutation swapping 1 and 2 and fixing 3 is

$$\tau = (1\ 2) = (2\ 1)$$

Multiplying cycles is not entirely trivial, but one can easily become proficient in the art. The most important thing is to start from the right.

Example 7.3

- $$\begin{aligned} \tau\sigma &= (1\ 2)(1\ 3\ 2) \\ &= (1\ 3) \end{aligned}$$

- $$(1\ 4\ 3\ 2)(2\ 4\ 3) = (1\ 4\ 2\ 3)$$

Remark. Cycles which are cyclic permutations of one another are the same: $(a_1\ \dots\ a_k) = (a_2\ \dots\ a_k\ a_1) = \cdots$

Definition 7.4. Cycles $(a_1\ \dots\ a_k)$ and $(b_1\ \dots\ b_l)$ are *disjoint* if they are disjoint as sets – $a_i \neq b_j$ for any i, j. Disjoint cycles have the nice property that they commute.

Theorem 7.5 (Disjoint cycles)

Every $\sigma \in S_n$ can be written as a product of disjoint cycles. Furthermore, this product is unique up to cyclic permutation of cycle elements and cycle reordering.

Proof. The action of $\langle\sigma\rangle$ on $X = \{1, \dots, n\}$ partitions X into orbits:

$$X = \langle\sigma\rangle\, i_1 \cup \langle\sigma\rangle\, i_2 \cup \cdots \cup \langle\sigma\rangle\, i_k$$

Let $n_j = |\langle\sigma\rangle\, i_j|$. Then

$$\sigma = (i_1\ \sigma(i_j)\ \dots\ \sigma^{n_1-1}(i_1)) \dots (i_k\ \sigma(i_k)\ \dots\ \sigma^{n_k-1}(i_k))$$

which proves existence. For uniqueness, note that the only choices we made were the orbit representatives in each orbit and the indexing of the orbits/their representatives. These correspond to cyclically permuting the cycles of σ and changing the order of the cycles respectively, which are exactly the failures of uniqueness that we are allowed to have. ∎

Definition 7.6 (Cycle types)**.** If $\sigma = (a_1^1 \ \dots \ a_{k_1}^1) \dots (a_1^l \ \dots \ a_{k_l}^l)$ is a product of disjoint cycles then σ is called a $(k_1, \dots, k_l)$-*cycle*, and $\{k_1, \dots, k_l\}$ is called σ's *cycle type*.

> **Remark.** If σ is a k-cycle, then σ has order k. Furthermore, if σ has cycle type $\{k_1, \dots, k_l\}$, then σ has order $\operatorname{lcm}(k_1, \dots, k_l)$.

Definition 7.7. A *transposition* is just another name for a 2-cycle.

Theorem 7.8 (Generation by transpositions)

The set of transpositions in S_n generates S_n.

Proof. We proceed by induction on n. In the base case, $n = 2$, and the result is obvious. For the inductive step, suppose S_{n-1} is generated by transpositions, and let $\sigma \in S_n$. If $\sigma(n) = n$, then $\sigma \in S_{n-1} \leq S_n$, and so we conclude by the inductive hypothesis. Otherwise, let $\tau = (n \ \sigma(n))$, so $\tau\sigma(n) = n$. Then, as before, it follows that $\tau\sigma \in S_{n-1}$, so $\tau\sigma$ is a product of transpositions – multiplying by $\tau^{-1} = \tau$ on both sides, σ is a product of transpositions as required. ■

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Lemma 7.9 (Adjacent transpositions)

Any transposition $(i \ j)$ can be written as a product of an odd number of adjacent transpositions.

Proof. Suppose $j > i$, we proceed by induction on $j - i$. In the base case, $j = i + 1$, and so $(i \ j) = (i \ i+1)$ as required. Otherwise, if $j > i + 1$, we write

$$(i \ j) = (j-1 \ j)(i \ j-1)(j-1 \ j)$$

and the result follows by the inductive hypothesis applied to $(i \ j-1)$. ■

So it turns out that S_n is generated by adjacent transpositions. But the really important part of this is the notion of parity.

Lemma 7.10

If $\tau_1, \dots, \tau_k$ are all transpositions, and

$$\sigma = \tau_1 \cdots \tau_k = e$$

then k is even.

Definition 7.11. An unordered pair $\{i, j\} \subseteq \{1, \dots, n\}$ is called an *inversion* of a permutation σ if $i < j$ but $\sigma(i) > \sigma(j)$.

Proof. By lemma 7.9, we may assume all τ_i are adjacent. We claim that, for any $\sigma = \tau_1 \cdots \tau_k$, $k \equiv$ the number of inversions of $\sigma \pmod 2$. We induct on k.

In the base case, $k = 0$, so $\sigma = e$ and we're done.

For the inductive step, let $\sigma = \tau\sigma'$, where $\tau = \tau_1 = (l\; l+1)$ and $\sigma' = \tau_2 \cdots \tau_k$. Which pairs $\{i, j\}$ could be inversions of σ' but not of σ, or vice versa? The only such pair would have to satisfy

$$\sigma'(i) = l\sigma'(j) = l + 1$$

since τ fixes everything other than l and $l + 1$. Thus

$$\begin{aligned} \sigma'(i) &= l < l + 1 = \sigma'(j) \\ \sigma(i) &= l + 1 > l = \sigma(j) \end{aligned}$$

So, if $i < j$, then $\{i, j\}$ is an inversion for σ but not for σ', while if $i > j$, then $\{i, j\}$ is an inversion for σ' but not for σ. In either case,

$$\begin{aligned} \#\text{inversions of } \sigma &= \#\text{inversions of } \sigma' \pm 1 \\ &\equiv \#\text{inversions of } \sigma' + 1 \pmod 2 \end{aligned}$$

as required. Then, the lemma follows, since e has 0 inversions. ∎

This enables us to define the *sign homomorphism.*

Theorem 7.12 (Sign homomorphism)
The map

$$\begin{aligned} \text{sign} : S_n &\to C_2 = \{\pm 1\} \\ \tau_1 \cdots \tau_k &\mapsto (-1)^k \end{aligned}$$

where τ_i are transpositions, is a well-defined homomorphism.

Proof.
- *Well-defined.* Suppose we have

$$\begin{aligned} &\tau_1 \cdots \tau_k = \tau_1' \cdots \tau_l' \\ \implies &\tau_k \cdots \tau_1 \tau_1' \cdots \tau_l' = e \end{aligned}$$

  since all the transpositions are self-inverse. Hence, by the lemma, $k + l$ is even, so $k \equiv l \pmod 2$, and so $(-1)^k = (-1)^l$ as required.

- *Homomorphism.* Clearly

$$\begin{aligned} \text{sign}(\tau_1 \cdots \tau_k \tau_1' \cdots \tau_l') &= (-1)^{k+l} \\ &= (-1)^k (-1)^l \\ &= \text{sign}(\tau_1 \cdots \tau_k)\,\text{sign}(\tau_1' \cdots \tau_l') \end{aligned}$$

∎

Definition 7.13 (Parity and the alternating group)**.** If $\text{sign}(\sigma) = 1$, then σ is called *even*, *odd* otherwise. The subgroup

$$A_n = \ker(\text{sign}) \lhd S_n$$

is called the *alternating group* on n elements – it is the group of even permutations.

Example 7.14

Let's find A_3. We know

$$S_3 = \{e, (1\ 2), (2\ 3), (1\ 3), (1\ 2\ 3), (3\ 2\ 1)\}$$

and we have $(1\ 2\ 3) = (1\ 2)(2\ 3)$, and $(3\ 2\ 1) = (2\ 3)(1\ 2)$, so these are even. Thus,

$$A_3 = \{e, (1\ 2\ 3), (3\ 2\ 1)\} \cong C_3$$

Remark. The cycle type makes it easy to determine a permutation's sign. Indeed,

$$(a_1\ \cdots\ a_k) = (a_1\ a_k)(a_1\ a_{k-1}) \cdots (a_1\ a_3)(a_1\ a_2)$$

so $(a_1\ \cdots\ a_k)$ is a product of $k-1$ transpositions. Thus, $(a_1\ \cdots\ a_k)$ is even if and only if k is odd. More generally, a $(k_1, \ldots, k_l)$-cycle is even if and only if $|\{k_i : k_i \text{ even}\}|$ is even. For example, $(1\ 2)(3\ 4)$ is even, but $(1\ 2)(3\ 4)(5\ 6)$ is odd.

Let's apply this to study conjugacy in S_n and A_n. S_n is surprisingly easy!

Theorem 7.15 (Conjugacy in S_n)

Two permutations σ_1, $\sigma_2 \in S_n$ are conjugate if and only if they have the same cycle type.

Proof. $\implies$ Suppose $\sigma_1 = (a_1^1\ \cdots\ a_{l_1}^1) \cdots (a_1^k\ \cdots a_{l_k}^k)$ is a product of disjoint cycles. Thus $\sigma_1(a_j^i) = a_{j+1}^i$, where the addition in the subscript is modulo l_i. Since the cycle types are the same, $\sigma_2 = (b_1^1\ \cdots\ b_{l_1}^1) \cdots (b_1^k\ \cdots b_{l_k}^k)$. Now consider the permutation τ given by $\tau(a_j^i) = b_j^i$. We have

$$\begin{aligned}
\tau\sigma_1\tau^{-1}(b_j^i) &= \tau\sigma_1(a_j^i) \\
&= \tau(a_{j+1}^i) \\
&= b_{j+1}^i \\
&= \sigma_2(b_j^i)
\end{aligned}$$

Therefore, $\sigma_2 = \tau\sigma_1\tau^{-1}$, so they are conjugate as required.

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$\impliedby$ Suppose $\sigma_2 = \tau\sigma_1\tau^{-1}$. The above argument shows that, if

$$\sigma_1 = (a_1^1\ \cdots\ a_{l_1}^1) \cdots (a_1^k\ \cdots a_{l_k}^k)$$

then

$$\sigma_2 = (b_1^1\ \cdots\ b_{l_1}^1) \cdots (b_1^k\ \cdots b_{l_k}^k)$$

where $b_j^i = \tau(a_j^i)$ for all relevant i, j. Therefore, σ_1 and σ_2 have the same cycle type.

∎

This makes it easy to count conjugacy classes in S_n.

Example 7.16 ($n = 3$)
Recall

$$S_3 = \{e, (1\ 2), (2\ 3), (1\ 3), (1\ 2\ 3), (3\ 2\ 1)\}$$

- $|\operatorname{ccl}_{S_3}(1\ 2)| = \binom{3}{2} = 3$;
- $|\operatorname{ccl}_{S_3}(1\ 2\ 3)| = 2 \cdot 1 = 2$.

Let's also recall that corollary 3.7 (the orbit-stabiliser theorem) implies that

$$|C_{S_n}(\gamma)| = \frac{|S_n|}{|\operatorname{ccl}_{S_n}(\gamma)|} = \frac{n!}{|\operatorname{ccl}_{S_n}(\gamma)|}$$

so we can also find the size of the centraliser pretty easily!

Example 7.17 ($n = 4$)
Now we want to do a similar calculation on the less familiar S_4.

- $|\operatorname{ccl}_{S_4}[(1\ 2)(3\ 4)]| = \frac{1}{2}\binom{4}{2} = 3$;
- $|C_{S_4}[(1\ 2)(3\ 4)]| = 24/3 = 8$. The cool thing about this is that we can start listing elements of the centraliser, and we know that, once we have 8 things, we're done. In particular, we have

$$\begin{aligned} C_{S_4}[(1\ 2)(3\ 4)] = \{&e, (1\ 2)(3\ 4), (1\ 2), (3\ 4), (1\ 3)(2\ 4), \\ &(1\ 4)(2\ 3), (1\ 4\ 2\ 3), (1\ 3\ 2\ 4)\} \end{aligned}$$

Let's do this in full generality.

Typical element γ	$\lvert\operatorname{ccl}_{S_4}(\gamma)\rvert$	$\lvert C_{S_4}(\gamma)\rvert$
e	1	24
$(1\ 2)$	$\binom{4}{2} = 6$	4
$(1\ 2)(3\ 4)$	$\frac{1}{2}\binom{4}{2} = 3$	8
$(1\ 2\ 3)$	$2 \cdot \binom{4}{3} = 8$	3
$(1\ 2\ 3\ 4)$	$3! = 6$	4

Counting conjugacy classes in A_n can also be done with a little thought.

Lemma 7.18 (Conjugacy classes in A_n)
Let $\gamma \in A_n \lhd S_n$.

- If an odd element of S_n commutes with γ, then $\operatorname{ccl}_{A_n}(\gamma) = \operatorname{ccl}_{S_n}(\gamma)$.
- If every element of S_n that commutes with γ is even, then $\operatorname{ccl}_{S_n}(\gamma)$ splits into two:

$$\operatorname{ccl}_{S_n}(\gamma) = \operatorname{ccl}_{A_n}(\gamma) \cup \operatorname{ccl}_{A_n}(\tau\gamma\tau^{-1})$$

where τ is any transposition.

Proof. By corollary 3.7 (orbit-stabiliser),

$$|S_n| = |\operatorname{ccl}_{S_n}(\gamma)||C_{S_n}(\gamma)|$$
$$|A_n| = |\operatorname{ccl}_{A_n}(\gamma)||C_{A_n}(\gamma)|$$

Then, since $|S_n| = 2|A_n|$, we have

$$|\operatorname{ccl}_{S_n}(\gamma)| = 2\frac{|C_{A_n}(\gamma)|}{|C_{S_n}(\gamma)|} \cdot |\operatorname{ccl}_{A_n}(\gamma)| \tag{$*$}$$

Now

$$C_{A_n}(\gamma) = \ker(\operatorname{sign}|_{C_{S_n}(\gamma)} : C_{S_n}(\gamma) \to C_2)$$

The image has size 1 or 2, so, by theorem 6.8 (the isomorphism theorem),

$$|C_{S_n(\gamma)} : C_{A_n(\gamma)}| = 1 \text{ or } 2$$

- If there is an odd element of S_n that commmutes with γ, we hit $-1 \in C_2$, so the image has size 2 and hence

  $$|C_{S_n(\gamma)} : C_{A_n(\gamma)}| = 2$$

  Then, Lagrange and $(*)$ gives $|\operatorname{ccl}_{S_n}(\gamma)| = |\operatorname{ccl}_{A_n}(\gamma)|$ as required.

- If no odd elements of S_n commute with γ, $|C_{S_n(\gamma)} : C_{A_n(\gamma)}| = 1$, $C_{S_n}(\gamma) = C_{A_n}(\gamma)$, so $(*)$ gives $|\operatorname{ccl}_{S_n}(\gamma)| = 2|\operatorname{ccl}_{A_n}(\gamma)|$, so $\operatorname{ccl}_{A_n}(\gamma)$ is half as big. Now take a transposition $\tau \in S_n$. Observe $\tau\gamma\tau^{-1} \in \operatorname{ccl}_{S_n}(\gamma)$. If $\tau\gamma\tau^{-1} \in \operatorname{ccl}_{A_n}(\gamma)$, then $\tau\gamma\tau^{-1} = \alpha\gamma\alpha^{-1}$ for some $\alpha \in A_n$. But then $(\tau\alpha)\gamma(\tau\alpha)^{-1} = \gamma$, while $\tau\alpha$ is odd, being the product of an odd τ and an even α, so $C_{S_n}(\gamma)$ contains an odd element. But this is a contradiction. Hence $\tau\gamma\tau^{-1} \notin \operatorname{ccl}_{A_n}(\gamma)$. From this it's fairly clear that

  $$\operatorname{ccl}_{S_n}(\gamma) = \operatorname{ccl}_{A_n}(\gamma) \cup \operatorname{ccl}_{A_n}(\tau\gamma\tau^{-1})$$

  as required.

■

Example 7.19 (A_4)

The even elements of S_4 are e, the $(2,2)$-cycles and the 3-cycles.

- Since there are an odd number of $(2,2)$-cycles, the conjugacy class of them cannot possibly split, so it remains intact.
- For a 3-cycle $\sigma = (1\ 2\ 3)$, $|C_{S_4}(1\ 2\ 3)| = 3$, so

$$C_{S_4}(1\ 2\ 3) = \langle (1\ 2\ 3) \rangle \leq A_4$$

so every element that commutes with $C_{S_4}(1\ 2\ 3)$ is even, and hence it splits.

Let us summarise:

Typical element γ	$\lvert \mathrm{ccl}_{A_4}(\gamma) \rvert$
e	1
$(1\ 2)(3\ 4)$	3
$(1\ 2\ 3)$	4
$(3\ 2\ 1)$	4

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Finally, let's look at conjugacy in S_5 and A_5.

Example 7.20 (Conjugacy in S_5)

We take the same approach as S_4.

Typical element γ	$\lvert \mathrm{ccl}_{S_5}(\gamma) \rvert$	$\lvert C_{S_5}(\gamma) \rvert$	sign
e	1	120	1
$(1\ 2)$	$\binom{5}{2} = 10$	12	-1
$(1\ 2\ 3)$	$2 \cdot \binom{5}{3} = 20$	6	1
$(1\ 2)(3\ 4)$	$\frac{1}{2}\binom{5}{2}\binom{3}{2} = 15$	8	1
$(1\ 2\ 3)(4\ 5)$	$2\binom{5}{3} = 20$	6	-1
$(1\ 2\ 3\ 4)$	$3! \cdot \binom{5}{4} = 30$	4	-1
$(1\ 2\ 3\ 4\ 5)$	$4! = 24$	5	1

Example 7.21 (Conjugacy in A_5)

- The transposition $(4\ 5)$ commutes with $(1\ 2\ 3)$ but is odd, so the conjugacy class doesn't split – $\mathrm{ccl}_{A_5}(1\ 2\ 3) = \mathrm{ccl}_{S_5}(1\ 2\ 3)$.
- There are an odd number of double transpositions, so they definitely can't split – $\mathrm{ccl}_{A_5}[(1\ 2)(3\ 4)] = \mathrm{ccl}_{S_5}[(1\ 2)(3\ 4)]$
- For 5-cycles, the centraliser is only of size 5 and, since it's of order 5, the elements of the centraliser must be exactly the cyclic subgroup it generates – $C_{S_5}(1\ 2\ 3\ 4\ 5) = \langle(1\ 2\ 3\ 4\ 5)\rangle \leq A_5$. Hence, only even permutations commute with $(1\ 2\ 3\ 4\ 5)$, and so the conjugacy class splits.

In summary,

Typical element γ	$\lvert \mathrm{ccl}_{A_5}(\gamma)\rvert$
e	1
$(1\ 2\ 3)$	20
$(1\ 2)(3\ 4)$	15
$(1\ 2\ 3\ 4\ 5)$	12
$(2\ 1\ 3\ 4\ 5)$	12

Theorem 7.22 (A_5 is simple)
A_5 is simple.

Proof. Suppose $N \triangleleft A_5$. Then N is a union of conjugacy classes in A_5. Hence, it is necessary to express a factor of 60 (by Lagrange) as a sum of the sizes of the conjugacy classes, and importantly we must have $\{e\}$ as one of our conjugacy classes. One can see that this cannot be done unless $N = \{e\}$ or $N = A_5$. ∎

§8 Matrix groups

Let $M_n(\mathbb{R}) = \{n$ by n matrices with entries in $\mathbb{R}\}$. Matrix multiplication is associative, and the identity matrix provides an identity. However, not all matrices are invertible.

Lemma 8.1
$A \in M_n(\mathbb{R})$ has an inverse if and only if $\det A \neq 0$.

Now we actually can make our matrices into a group.

Definition 8.2 (General linear group)**.** Let $\mathrm{GL}_n(\mathbb{R}) = \{A \in M_n(\mathbb{R}) : \det A \neq 0\}$. This is a group under matrix multiplication by the above, namely the *general linear group*.

Lemma 8.3
For $A, B \in M_n(\mathbb{R})$, $\det(AB) = \det A \det B$.

Corollary 8.4

It follows that det is a homomorphism:

$$\begin{aligned} \det : \mathrm{GL}_n(\mathbb{R}) &\to \mathbb{R}_\times = (\mathbb{R} \setminus \{0\}, 1, \times) \\ A &\mapsto \det A \end{aligned}$$

Definition 8.5 (Special linear group)**.** The *special linear group* $\mathrm{SL}_n(\mathbb{R}) = \ker(\det) = \{A \in M_n(\mathbb{R}) : \det A = 1\}$.

By theorem 6.8 (the isomorphism theorem), $\mathrm{SL}_n(\mathbb{R}) \triangleleft \mathrm{GL}_n(\mathbb{R})$, and $\mathrm{GL}_n(\mathbb{R}) / \mathrm{SL}_n(\mathbb{R}) \cong \mathrm{im}(\det) \cong \mathbb{R}_\times$ – it's fairly clear to see that any real number can arise as a determinant.

Furthermore, everything above still works with $\mathbb{C}$, so we can talk about $\mathrm{GL}_n(\mathbb{C})$, $\mathrm{SL}_n(\mathbb{C})$, and analogously we have $\mathrm{GL}_n(\mathbb{C}) / \mathrm{SL}_n(\mathbb{C}) \cong \mathbb{C}_\times$.

§8.1 Change of basis

There is a natural action by conjugation of $\mathrm{GL}_n(\mathbb{R})$ on $M_n(\mathbb{R})$:

$$\begin{aligned} \mathrm{GL}_n(\mathbb{R}) &\curvearrowright M_n(\mathbb{R}) \\ P(A) &:= PAP^{-1} \end{aligned}$$

Proposition 8.6

Let V be an n-dimensional vector space over $\mathbb{R}$, and $\alpha : V \to V$ a linear map. If $A \in M_n(\mathbb{R})$ represents α in some basis, then the orbit

$$\mathrm{GL}_n(\mathbb{R})A = \{PAP^{-1} : P \in \mathrm{GL}_n(\mathbb{R})\}$$

consists of all the matrices representing α in any basis.

Proof. A basis $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ for V defines an isomorphism of vector spaces

$$\begin{aligned} \phi : \mathbb{R}^n &\to V \\ (\lambda_1, \dots, \lambda_n) &\mapsto \sum_{i=1}^{n} \lambda_i \mathbf{v}_i \end{aligned}$$

The claim that A represents α in this basis means that the following diagram commutes

$$\begin{tikzcd} \mathbb{R}^n \arrow[r, "\phi"] \arrow[d, "A"'] & V \arrow[d, "\alpha"] \\ \mathbb{R}^n \arrow[r, "\phi"'] & V \end{tikzcd}$$

– that is, $\alpha = \phi A \phi^{-1}$.

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Likewise, another basis $\{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ for V corresponds to another isomorphism $\psi : \mathbb{R}^n \to V$, and another matrix B representing α in these coordinates, which again satisfies $\alpha = \psi B \psi^{-1}$. Therefore,

$$\begin{aligned} B &= \psi^{-1}\alpha\psi \\ &= (\psi^{-1} \circ \phi)A(\phi^{-1} \circ \psi) \\ &= PAP^{-1} \end{aligned}$$

where $P \in \mathrm{GL}_n(\mathbb{R})$ represents the linear map $\psi^{-1} \circ \phi : \mathbb{R}^n \to \mathbb{R}^n$ in the standard basis (note we know P is invertible since P^{-1} is the matrix representing $\phi^{-1} \circ \psi$). Thus, the set of all matrices representing α is contained in the orbit of A under conjugation.

Conversely, if $B = PAP^{-1}$ for some $P \in \mathrm{GL}_n(\mathbb{R})$, putting $\psi = \phi \circ P^{-1} : \mathbb{R}^n \to V$ gives us a basis $\{\mathbf{u}_i = \psi(\mathbf{e}_i)\}$ for V. In this basis, B represents α. ■

§8.2 Möbius transformations: a new perspective

Recall that multiplication in $\mathcal{M}$ looks similar to multiplication of 2 by 2 matrices.

Proposition 8.7

Represent $\mathbb{C}_\times$ as a matrix group by

$$\mathbb{C}_\times = \left\{ \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \in \mathrm{GL}_2(\mathbb{C}) : \lambda \in \mathbb{C}_\times \right\}$$

Then $\mathbb{C}_\times \lhd \mathrm{GL}_2(\mathbb{C})$, and $\mathcal{M} \cong \mathrm{GL}_2(\mathbb{C})/\mathbb{C}_\times$

Proof. Let

$$\begin{aligned} \Phi : \mathrm{GL}_2(\mathbb{C}) &\to \mathcal{M} \\ \begin{pmatrix} a & b \\ c & d \end{pmatrix} &\mapsto \left(z \mapsto \frac{az+b}{cz+d} \right) \end{aligned}$$

By our computation of multiplication in $\mathcal{M}$, Φ is a homomorphism, and it's clearly surjective, so $\operatorname{im}\Phi = \mathcal{M}$. Furthermore, a matrix M is in $\ker\Phi$ if and only if its image fixes 0, 1 and ∞ Then

- M fixes 0 $\implies b = 0$;
- M fixes 1 $\implies a = d$;
- M fixes ∞ $\implies c = 0$.

So $\ker\Phi = \mathbb{C}_\times$. The result now follows by the isomorphism theorem. ■

> **Remark.** We get to simultaneously prove that $\mathbb{C}_\times$ is normal and identify the quotient all just by looking at our homomorphism.

§8.3 Orthogonal groups

Let's write $||\cdot||$ for the usual notion of length in $\mathbb{R}^n$ (the Euclidean norm) – that is,

$$||\mathbf{u}|| = \sqrt{\sum_{i=1}^{n} u_i^2}$$

Definition 8.8. The *n-dimensional orthogonal group*, $O(n)$, is the subgroup of $\mathrm{GL}_n(\mathbb{R})$ that preserves the Euclidean norm.

In fact, the dot product is often more convenient to work with:

$$\mathbf{u}\cdot\mathbf{v} = \sum_{i=1}^{n} u_i v_i$$

and it turns out that matrices preserve all dot products if and only if they preserve dot products of vectors with themselves, i.e. the Euclidean norm.

Lemma 8.9 (Polarisation identity)
For $\mathbf{u}$, $\mathbf{v} \in \mathbb{R}^n$,

$$2(\mathbf{u}\cdot\mathbf{v}) = ||\mathbf{u}||^2 + ||\mathbf{v}||^2 - ||\mathbf{u}-\mathbf{v}||^2$$

Proof. Observe

$$\begin{aligned} ||\mathbf{u}-\mathbf{v}||^2 &= (\mathbf{u}-\mathbf{v})\cdot(\mathbf{u}-\mathbf{v}) \\ &= ||\mathbf{u}||^2 - 2(\mathbf{u}\cdot\mathbf{v}) + ||\mathbf{v}||^2 \end{aligned}$$

whence the identity follows by rearrangement. ■

In other words, length-preserving transformations also preserve angles. Importantly for us, this allows us to characterise $O(n)$ using the dot product.

Lemma 8.10 ($O(n)$ and the dot product)
$O(n) = \{A \in \mathrm{GL}_n(\mathbb{R}) : (A\mathbf{x})\cdot(A\mathbf{y}) = \mathbf{x}\cdot\mathbf{y}\forall \mathbf{x},\mathbf{y} \in \mathbb{R}^n\}$.

Proof. If $(A\mathbf{x})\cdot(A\mathbf{y}) = \mathbf{x}\cdot\mathbf{y}$ for all $\mathbf{x}$, $\mathbf{y} \in \mathbb{R}^n$, then, for any $\mathbf{v} \in \mathbb{R}^n$,

$$\begin{aligned} ||A\mathbf{v}||^2 &= (A\mathbf{v})\cdot(A\mathbf{v}) \\ &= \mathbf{v}\cdot\mathbf{v} = ||\mathbf{v}||^2 \end{aligned}$$

We may take square roots on both sides since the norm is positive. Conversely, if $A \in O(n)$, then, for any $\mathbf{x}$, $\mathbf{y} \in \mathbb{R}^n$,

$$\begin{aligned} 2(A\mathbf{x})\cdot(A\mathbf{y}) &= ||A\mathbf{x}||^2 + ||A\mathbf{y}||^2 - ||A\mathbf{x} - A\mathbf{y}||^2 \\ &= ||\mathbf{x}||^2 + ||\mathbf{y}||^2 + ||\mathbf{x}-\mathbf{y}||^2 \\ &= 2\mathbf{x}\cdot\mathbf{y} \end{aligned}$$

■

This quickly leads to a nice characterisation of matrices in $O(n)$.

Lemma 8.11 (Matrices in $O(n)$)

Let $A \in M_n(\mathbb{R})$. The following are equivalent:

① $A \in O(n)$;

② the columns of A form an orthonormal basis;

③ $A^\intercal A = I$.

Proof. Let $A = (a_{ij})$.

(1 $\implies$ 2) Let $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ be the standard basis for $\mathbb{R}^n$. The i^{th} column of A is the column vector $A\mathbf{e}_i$. But then

$$(A\mathbf{e}_i) \cdot (A\mathbf{e}_j) = \mathbf{e}_i \cdot \mathbf{e}_j = \delta_{ij}$$

so the columns form an orthonormal basis, as required.

(2 $\implies$ 3) We know $(A\mathbf{e}_i) \cdot (A\mathbf{e}_j) = \delta_{ij}$. Then

$$\begin{aligned}(A^\intercal A)_i j &= \mathbf{e}_i^\intercal (A^\intercal A)\mathbf{e}_j \\ &= (A\mathbf{e}_i)^\intercal (A\mathbf{e}_j) = \delta_{ij}\end{aligned}$$

so $A^\intercal A = I$ as required.

(3 $\implies$ 1) Suppose $\mathbf{u}$, $\mathbf{v} \in \mathbb{R}^n$. Then

$$\begin{aligned}(A\mathbf{u}) \cdot (A\mathbf{v}) &= (A\mathbf{u})^\intercal (A\mathbf{v}) \\ &= \mathbf{u}^\intercal A^\intercal A\mathbf{v} \\ &= \mathbf{u}^\intercal \mathbf{v} \\ &= \mathbf{u} \cdot \mathbf{v}\end{aligned}$$

so $A \in O(n)$ as required.

■

Recall that $\det A^\intercal = \det A$. Therefore,

$$1 = \det I = \det(A^\intercal A) = (\det A)^2$$

and so the determinant of any orthogonal matrix is ± 1.

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Definition 8.12. The *special orthogonal group* $SO(n) = O(n) \cap SL_n(\mathbb{R})$ – the set of orthogonal matrices with determinant 1. Observe also that

$$SO(n) = \ker(\det : O(n) \to \{\pm 1\})$$

so $|O(n) : SO(n)| = 2$.

Examples of elements of $O(n)$ which are not in $SO(n)$ are provided by reflections.

Definition 8.13 (Reflections)**.** Any $\mathbf{v} \in \mathbb{R} \setminus \{0\}$ defines an *orthogonal plane*

$$\mathbf{v}^\perp = P_\mathbf{v} = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} \cdot \mathbf{v} = 0\}$$

The *reflection* in $P_\mathbf{v}$ is defined as

$$S_\mathbf{v}(\mathbf{x}) = \mathbf{x} - \frac{2(\mathbf{x} \cdot \mathbf{v})}{||\mathbf{v}||^2}\mathbf{v}$$

DIAGRAM: any diagram that makes this definition make sense.

Remark 8.14.

- We will sometimes write S_P for the reflection in the plane P.
- We may replace $\mathbf{v}$ by $\frac{\mathbf{v}}{||\mathbf{v}||}$ and so we can clearly assume $||\mathbf{v}|| = 1$ – then $S_\mathbf{v}(\mathbf{x}) = \mathbf{x} - 2(\mathbf{x} \cdot \mathbf{v})\mathbf{v}$.

Lemma 8.15

$S_{\hat{\mathbf{v}}}^2 = I$ for any $\hat{\mathbf{v}}$, and $S_{\hat{\mathbf{v}}} \in O(n)$.

Proof. Note that $S_{\hat{\mathbf{v}}}$ is linear in $\mathbf{x}$, so $S_{\hat{\mathbf{v}}} \in M_n(\mathbb{R})$. Now, observe

$$\begin{aligned} S_{\hat{\mathbf{v}}}(\mathbf{x}) \cdot \mathbf{v} &= (\mathbf{x} \cdot \hat{\mathbf{v}}) - 2(\mathbf{x} \cdot \hat{\mathbf{v}})(\hat{\mathbf{v}} \cdot \hat{\mathbf{v}}) \\ &= -\mathbf{x} \cdot \mathbf{v} \end{aligned}$$

Then

$$\begin{aligned} S_{\hat{\mathbf{v}}}^2(\mathbf{x}) &= S_{\hat{\mathbf{v}}}(\mathbf{x}) - 2(S_{\hat{\mathbf{v}}}(\mathbf{x}) \cdot \hat{\mathbf{v}})\hat{\mathbf{v}} \\ &= \mathbf{x} - 2(\mathbf{x} \cdot \hat{\mathbf{v}})\hat{\mathbf{v}} + 2(\mathbf{x} \cdot \hat{\mathbf{v}})\hat{\mathbf{v}} \\ &= \mathbf{x} \end{aligned}$$

so $S_{\hat{\mathbf{v}}}^2 = I$ as required. Hence $S_{\hat{\mathbf{v}}}$ is non-singular, so $S_{\hat{\mathbf{v}}} \in \mathrm{GL}_n(\mathbb{R})$. Finally, for $\mathbf{x} \in \mathbb{R}^n$,

$$\begin{aligned} ||S_{\hat{\mathbf{v}}}(\mathbf{x})||^2 &= (\mathbf{x} - 2(\mathbf{x} \cdot \hat{\mathbf{v}})\hat{\mathbf{v}})(\mathbf{x} - 2(\mathbf{x} \cdot \hat{\mathbf{v}})\hat{\mathbf{v}}) \\ &= ||\mathbf{x}||^2 - 4(\mathbf{x} \cdot \hat{\mathbf{v}})^2 + 4(\mathbf{x} \cdot \hat{\mathbf{v}})^2 \\ &= ||\mathbf{x}||^2 \end{aligned}$$

so $S_{\hat{\mathbf{v}}} \in O(n)$ as required. ∎

Remark 8.16. Take a basis $\{\mathbf{v}_1, \dots, \mathbf{v}_{n-1}\}$ for $P_{\hat{\mathbf{v}}}$. Then, in the basis $\{\mathbf{v}_1, \dots, \mathbf{v}_{n-1}, \hat{\mathbf{v}}\}$, $S_{\hat{\mathbf{v}}}$ has matrix

$$\begin{pmatrix} 1 & \cdots & 0 & 0 \\ \vdots & \ddots & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

so $\det S_{\hat{\mathbf{v}}} = 1$ and hence $S_{\hat{\mathbf{v}}} \notin SO(n)$.

Theorem 8.17 (Generation by reflections)

Every $A \in O(n)$ is a product of at most n reflections.

Proof. We proceed by induction on n.

Base case. $n = 1$. $O(1) = \{\pm 1\} = \langle -1 \rangle$.

Inductive step. Let $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ be the standard basis for $\mathbb{R}^n$. Let $\mathbf{v} = \mathbf{e}_n - A\mathbf{e}_n$.

DIAGRAM: $\mathbf{e}_n$ and $A\mathbf{e}_n$ shown as position vectors from the origin. They are connected, and perpendicular bisector drawn – this is the plane we reflect in.

Then $S_\mathbf{v} A\mathbf{e}_n = \mathbf{e}_n$. Since $S_\mathbf{v} A$ is orthogonal, it follows that $S_\mathbf{v} A$ preserves the plane $P_{\mathbf{e}_n} = \mathbb{R}^{n-1} \times \{0\}$. By induction, there exist $\mathbf{v}_1, \ldots, \mathbf{v}_{n-1} \in \mathbb{R}^{n-1}$ such that $S_\mathbf{v} A = S_{\mathbf{v}_1} \cdots S_{\mathbf{v}_{n-1}}$ on $\mathbb{R}^{n-1}$. Since both sides also fix $\mathbf{e}_n$, it follows they are also equal on $\mathbb{R}^n$. Hence,

$$A = S_\mathbf{v} S_{\mathbf{v}_1} \cdots S_{\mathbf{v}_{n-1}}$$

as required. ■

Let us consider orthogonal transformations in low dimensions – especially $n = 2$.

Definition 8.18 (Rotations)**.** A *rotation* is

- in 2 dimensions, an orthogonal matrix that only fixes the origin;
- in three dimensions, an orthogonal matrix that only fixes a line.

Lemma 8.19 (Elements of $O(2)$)

Let $A \in O(2)$. If $A \notin SO(2)$, then A is a reflection; if $A \in SO(2)$, then A is a rotation about the origin.

Proof. Recall that $\det S_\mathbf{v} = -1$, so the product of k reflections has determinant $(-1)^k$. By the theorem, we may take $k \leq 2$.

- If $A \notin SO(2)$, then k must be odd, so k must be 1, so A is a reflection.
- If $A \in SO(2)$, then k is even so, unless $A = I$, $A = S_\mathbf{u} S_\mathbf{v}$ for some $\mathbf{u}$, $\mathbf{v}$ not parallel. We claim that A only fixes $\mathbf{0}$. Suppose we have $\mathbf{x} \neq \mathbf{0}$ such that

  $$S_\mathbf{u} S_\mathbf{v} \mathbf{x} = \mathbf{x} \iff S_\mathbf{u} \mathbf{x} = S_\mathbf{v} \mathbf{x}$$

  But $\mathbf{v}$ is parallel to $\mathbf{x} - S_\mathbf{v}\mathbf{x}$, and $\mathbf{u}$ is parallel to $\mathbf{x} - S_\mathbf{u}\mathbf{x}$, so $\mathbf{u}$ and $\mathbf{v}$ are parallel, which is a contradiction. Hence A only fixes 0 as required, so it is a rotation.

■

Remark. We've defined rotations to be orthogonal transformations that only fix the origin. But we can show that this is equivalent to other definitions. For example, let $A \in SO(2)$, so the columns of A form an orthonormal basis. For the vectors to be perpendicular, we require

$$A = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$$

and then we further require $a^2 + b^2 = 1$. But then $a = \cos\theta$, $b = \sin\theta$ for some θ, so

$$A = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$$

as required.

That concludes our survey of $O(2)$ and $SO(2)$.

Lemma 8.20 (Elements of $SO(3)$)
If $A \in SO(3)$, A is a rotation.

Proof. As above, A is a product of at most 3 reflections so, for $\det A = 1$, $A = I$ or $A = S_{\mathbf{u}}S_{\mathbf{v}}$ for non-parallel $\mathbf{u}$, $\mathbf{v}$. It follows that

$$P_{\mathbf{u}} \cap P_{\mathbf{v}} = l$$

a line, so A fixes l. Also, $S_{\mathbf{u}}S_{\mathbf{v}}x = x \implies S_{\mathbf{u}}x = S_{\mathbf{v}}x$ and, arguing as in the previous lemma, we find that either $S_{\mathbf{u}} = S_{\mathbf{v}}$, in which case $A = I$, or $x \in S_{\mathbf{u}} \cap S_{\mathbf{v}}$. Hence A fixes only the line l, so A is a rotation. ■

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§9 Platonic solids

Amazingly, while there are infinitely many regular 2-dimensional polygons, there are only 5 *Platonic solids* in 3 dimensions.

Definition 9.1. A convex polyhedron $X \subseteq \mathbb{R}^3$ is a *Platonic solid* if

- every face is a regular n-gon for some constant n;
- $G = \mathrm{Isom}(X)$ acts transitively on the faces;
- if $x \in X$ is the midpoint of a face, then $\mathrm{Stab}_G(x) \cong D_{2n}$.

Let us consider the 5 solids.

- *The tetrahedron.* 4 triangular faces, 4 vertices.
- *The cube/hexahedron.* 6 square faces, 8 vertices.
- *The octahedron.* 8 triangular faces, 6 vertices.
- *The dodecahedron.* 12 pentagonal faces, 20 vertices.
- *The icosahedron.* 20 triangular faces, 12 vertices.

Definition 9.2. Two solids X and Y are *dual* if Y can be constructed from X by putting vertices in the centres of each face, and joining vertices in adjacent faces.

The cube and the octahedron are dual (DIAGRAM), as are the dodecahedron and icosahedron. The tetrahedron is dual to itself. Furthermore, if X and Y are dual, $\mathrm{Isom}(X) \cong \mathrm{Isom}(Y)$, so we only have three groups to identify!

Example 9.3 (Tetrahedron)

Let $G = \text{Isom}(\text{tetrahedron})$. We want to look for a group action. By definition, G acts transitively on the (4) faces, and each stabiliser is isomorphic to D_6. Thus, orbit-stabiliser gives

$$|G| = 4 \times 6 = 24$$

Furthermore, the action of G on the 4 vertices defines a homomorphism $\theta : G \to S_4$. Let us prove that θ is injective. Suppose $\theta(g) = e$. Then g fixes four vertices, which are not coplanar, and so $g = \text{id}$ by the four-point lemma, meaning θ is injective as required. Then, by comparing sizes, we see $G \cong S_4$.

Let's also identify the group of rotational symmetries:

$$G_0 = G \cap SO(3)$$

assuming our tetrahedron is centred on the origin.

Lemma 9.4 (Uniqueness of A_n)

If $H \leq S_n$ and $|S_n : H| = 2$, then $H = A_n$.

Proof. Because $|S_n : H| = 2$, $H \triangleleft S_n$ and $S_n/H \cong C_2$. We therefore have a surjective homomorphism $\theta : S_n \to C_2$ with $\ker\theta = H$. ■

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Finally, (non-examinable),

Example 9.5 (Dodecahedron/icosahedron)

Let $G = \text{Isom}(\text{dodecahedron})$ and G_0 the rotational subgroup of G (of index 2). Since a face has orbit of size 12 and stabiliser of size 10, $|G| = 120$, $|G_0| = 60$.

By drawing diagonals on the faces, we may inscribe 5 cubes into the dodecahedron. Since the cubes are built symmetrically from the geometry of the dodecahedron, G_0 acts on them, and so we obtain a homomorphism

$$\theta : G_0 \to S_5$$

The rotation around the axis through an opposite pair of vertices gives rise to a 3-cycle. There are ten different pairs of opposite vertices, so we get ten pairs of 3-cycles which are one another's inverses – that way, we get all 20 3-cycles in S_5.

We claim that A_5 is generated by 3-cycles. Since the 3-cycles form a conjugacy class, we know the subgroup generated by 3-cycles is normal in A_5 but, since A_5 is simple, it must be A_5 itself. Hence, $A_5 \leq \operatorname{im}\theta$, but

$$60 = |G_0| \geq |\operatorname{im}\theta| \geq |A_5| = 60$$

so θ is an isomorphism – $G_0 \cong A_5$.

Finally, just as for the cube, $-e$ is a symmetry of the dodecahedron which commutes with everything, so, by the direct product theorem, $G \cong A_5 \times C_2 \cong S_5$.